

Non-Invasive, Non-Contact Conductivity Measurement Using Radiofrequency Probe Loading

by

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Abstract

Radiofrequency probes are a vital component in magnetic resonance and magnetic resonance imaging instrumentation, with much of their foundational literature rooted in these fields. It is important to note that the work of this thesis is not magnetic resonance. Instead, this thesis explores a non-invasive, non-contact method for measuring conductivity in water-based samples using radiofrequency probe loading. We explore how these probes react to changing conductivities in brine solutions by tracking changes in their quality factor with a network analyzer. These quality factor shifts serve as proxies for inductive losses, directly tying conductivity to energy dissipation. Our tests span three probe designs: solenoidal probes and loop-gap resonators with cylindrical samples, and surface coils with nominally infinite planar samples. Theoretical predictions were compared with experimental results to understand how probe design and sample geometry affect conductivity measurements. We further investigated the impact of flowing versus stationary samples and tested different salts to assess broader applicability. Future work will refine this technique using real samples, advancing RF probe optimization and expanding its applications in material characterization and biological studies.

Acknowledgements

Each winter semester since my second year, I have found myself knocking on professors' doors, searching for research opportunities. It's an experience that is equal parts thrilling and nerve-wracking. But last winter was particularly tough. After a string of polite no's, I found myself in Dr. Ben's office, panting (perhaps more dramatically than necessary) about how hard it was to secure a summer research spot. Little did I know, a single email from him to the Physics faculty would set everything in motion. That email landed in Dr. Bruce's inbox, and soon after, I was introduced to this project. Dr. Bruce not only gave me the invaluable summer research opportunity that led to this honours project, he also mastered the fine balance between encouragement and scientific challenge. His insightful advice and unwavering support were instrumental in shaping the direction and outcome of this work. A huge thank you to Dr. Andrés, who played a crucial role in my initial experimental works. His patience and hands-on expertise turned complex setups into something almost (but not quite) foolproof. His generosity with both knowledge and lab time made all the difference. To everyone at the UNB MRI Research Centre - thank you for the stimulating discussions, occasional debates, and the collaborative spirit that made this research all the more exciting. Science is rarely a solo adventure, and I am grateful to have had such an inspiring group along for the ride.

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List of Symbols, Nomenclature, or Abbreviations

a	Probe radius
$\mathbf{A}$	Magnetic vector potential
AC	Alternating current
b	Sample radius
$\mathbf{B}$	Magnetic field
C	Capacitance
Δf	Frequency shift
EC	Electrochemical
$\mathbf{E}$	Electric field
ϵ_0	Permittivity of free space
ESR	Electron Spin Resonance
f_0	Resonant frequency
g	Coil half-length
I	Current
$\mathbf{J}$	Current density
L	Inductance
LC	Inductor-Capacitor
LGR	Loop-Gap Resonator
MR	Magnetic Resonance
MRI	Magnetic Resonance Imaging
μ_0	Permeability of free space
n	Number of coil turns
NMR	Nuclear Magnetic Resonance
Q	Quality factor (Q-factor)
R_{eff}	Effective resistance
R_m	Inductive losses
RF	Radiofrequency
σ	Conductivity
VNA	Vector Network Analyzer
ω_0	Angular resonant frequency

Chapter 1

Introduction

Electrical conductivity is a fundamental property of importance in many scientific and engineering applications, from assessing water quality to monitoring physiological changes in biological tissues [1, 2, 3]. Traditional methods for measuring conductivity, such as electrochemical (EC) sensors and toroidal conductivity meters, typically rely on direct sample interaction—either through electrode immersion or inductive coupling [2, 3, 4]. These approaches often require complex calibration to account for factors such as electrode degradation, sample geometry, and environmental conditions, constraining their adaptability in non-contact or dynamic applications [2, 3, 4].

We present a non-invasive, non-contact conductivity measurement technique using radiofrequency (RF) probe loading, which leverages inductive interactions between a resonant circuit and the sample. In magnetic resonance imaging (MRI), RF probe loading presents a challenge: the human body, from one perspective, is just a large bag of saltwater, with eddy currents that heat tissue. Add to that the fact that humans vary in shape and size, further complicates probe design and performance. Hoult and Lauterbur [5] demonstrated that noise in magnetic resonance (MR) experiments with conductive samples arises not only from the resistance of the receiving coil but also from dielectric and inductive losses within the sample. While proper

shielding and grounding [5, 6, 7] can minimize dielectric losses, no effective methods exist to mitigate inductive losses. Unlike MR applications of RF probes, our approach employs a simplified experimental setup that isolates inductive losses due to sample conductivity, exploiting these losses as a means to measure conductivity.

This study investigates how the quality factor (Q) of an RF circuit responds to varying sample conductivities. The shift in Q upon introducing a sample reveals its inductive losses, forging a direct link between those losses and conductivity. To evaluate this approach, we characterized and tested three distinct RF probe configurations: solenoidal coils and a loop-gap resonator (LGR) with cylindrical samples, and a surface coil with effectively infinite planar samples. The resonant behavior of each probe was analyzed using a network analyzer, enabling precise measurements of the Q -factor under different salinity conditions. The theoretical predictions of the RF probe behavior were derived from first-principles electromagnetism and compared to experimental outcomes. We further explored the method’s versatility by examining flowing versus stationary solution samples and assessing the influence of different salts and common food samples, broadening its potential applications.

The remainder of this thesis is structured as follows: Chapter 2 provides the theoretical background on RF probe loading, detailing the relationship between inductive losses and conductivity across various probe geometries. Chapter 3 outlines the characterization process for tuning, matching, and measuring the Q -factor of RF probes. Chapter 4 presents the experimental design, results, and analysis, including initial validation of the theory with saline solutions and exploratory tests with flowing fluids and food samples. Chapter 5 concludes with a summary of findings and directions for future research. These investigations lay the groundwork to refine this method into a promising alternative for conductivity measurements in water-based samples where traditional tools falter, opening doors to real-world uses like biological fluid characterization and material sensing.

Chapter 2

Theory

2.1 RF Probe Loading

RF probe loading is a non-contact method to measure the conductivity of water-based samples using electromagnetic interactions. The concept is simple: an RF signal – essentially an alternating current (AC) oscillating at radio frequencies – flows through the RF probe, typically a coil that forms part of a resonant circuit designed to transmit and receive RF signals. This current generates an oscillating magnetic field extending into the surrounding space, which, when a conductive sample is placed nearby, induces eddy currents within the sample, altering the probe’s electromagnetic response. A simple schematic illustrating this concept is presented in Fig. 2.1, where a surface coil serves as the probe interacting with an adjacent planar conductive sample.

Consider the RF probe in action: current oscillates at a radiofrequency, creating an oscillating magnetic field described by Ampere-Maxwell’s law ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$), where the field’s curl is primarily driven by the current density ($\mathbf{J}$), with the displacement current term ($\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$) playing a minor role at these frequencies [8, 9, 10]. When a conductive sample enters this field, Faraday’s law ($\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$) dictates that the changing magnetic field ($\partial \mathbf{B} / \partial t$) induces an electric field

($\mathbf{E}$) within the sample. In a water-based medium rich with charged ions, this electric field mobilizes charges, generating eddy currents – circulating loops of current that traverse the sample [5, 9, 10, 11, 12, 13].

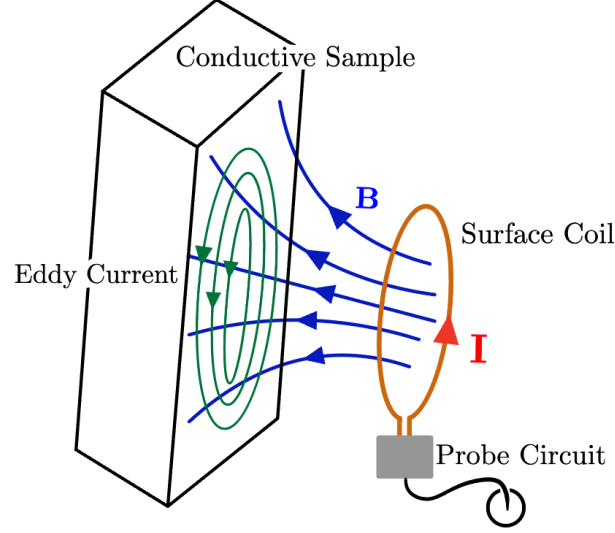


Figure 2.1: A simple schematic representation of a surface coil RF probe interacting with a conductive sample. An alternating current ($\mathbf{I}$, shown in red) in the coil generates a magnetic field ($\mathbf{B}$, shown in blue), which induces eddy currents (green loop) in the conductive sample. These eddy currents dissipate energy through resistive heating in the sample, increasing the RF circuit’s effective resistance and altering its resonance.

These eddy currents are the linchpin of our measurement. Governed by Ohm’s law in the sample ($\mathbf{J} = \sigma \mathbf{E}$, where σ is conductivity), their strength scales with conductivity: the more conductive the sample, the more pronounced the currents. The eddy currents also depend on the frequency of the alternating magnetic field, as higher frequencies enhance the rate of change ($\partial \mathbf{B} / \partial t$), amplifying the induced field and thus the currents. These currents dissipate energy as heat through Joule heating ($P = J^2 / \sigma$ per unit volume) [5]. Often referred to as inductive losses or eddy current losses, R_m , this energy dissipation reflects the sample’s ability to absorb and dissipate electromagnetic energy, a phenomenon that underpins our measurement approach.

This process is deeply rooted in Maxwell’s equations, where the sample’s eddy currents couple to the probe via mutual inductance, subtly altering the circuit’s electrical properties [8]. The extent of these losses induced by the sample provides a measurable signature of its conductivity, setting the stage for detailed analysis. With this foundation in place, we turn to explore how these losses manifest in the probe’s resonance.

2.1.1 Quality Factor

The quality factor, or Q-factor, of the resonant circuit stands at the heart of RF probe loading, providing insight into how a probe’s resonance changes in the presence of a conductive sample. This dimensionless parameter gauges the efficiency of a resonant circuit by comparing the energy it stores to the energy it dissipates over one cycle of the RF signal [7, 8, 11, 12, 14]. It is formally defined as:

$$Q = \frac{\text{Maximum Energy Stored}}{\text{Average Energy Dissipated per Cycle}} = \frac{\omega_0 L}{R_{\text{eff}}}, \quad (2.1)$$

where $\omega_0 = 2\pi f_0$ is the resonant frequency, L is the inductance, and R_{eff} is the effective resistance, encompassing both intrinsic circuit losses and those induced by the sample. A high Q-factor signals a sharp resonance with minimal energy loss, while a lower Q-factor indicates broader resonance due to increased dissipation (see Figure 3.4). In RF probe loading, these changes in Q-factor directly reflect the electromagnetic interactions between the probe and the sample, making it fundamental to our conductivity measurements.

2.1.2 Losses

The effective resistance, R_{eff} encapsulates all sources of energy loss in the RF probe and is expressed as:

$$R_{\text{eff}} = R_{\text{coil}} + R_{\text{dielectric}} + R_{\text{radiative}} + R_{\text{m}}, \quad (2.2)$$

where each term reflects distinct physical mechanisms. The resistive loss, R_{coil} , arises from ohmic resistance in the probe's wires, amplified at high frequencies by the skin effect, which confines current to the conductor's surface [8]. To minimize this, low-resistance materials like copper are employed. Dielectric losses, $R_{\text{dielectric}}$, stem from electric fields fringing into the sample due to potential differences across the circuit, interacting with lossy dielectric materials [12]. In saline solutions, however, these are typically overshadowed by inductive losses, R_{m} , driven by eddy currents, as the sample's conductivity dominates [11]. Radiative losses, $R_{\text{radiative}}$, occur at higher frequencies [14] but remain minor in our setup compared to resistive and inductive contributions.

Each term in Eq. 2.2 depends on the probe's design – its geometry and materials – and the operating frequency. Ideally, a probe minimizes all these losses, but in this study, the design prioritizes the sample's contribution, such that $R_{\text{eff}} \approx R_{\text{m}}$. This simplifies the Q-factor to $Q \approx \omega_0 L / R_{\text{m}}$, directly linking it to the inductive losses.

2.1.3 Relating Inductive Losses to Q-factor

A natural question thus arises: does our approach sacrifice accuracy by assuming that inductive losses dominate, such that $R_{\text{eff}} \approx R_{\text{m}}$? This simplification, while an idealization, serves to highlight the optimal case where sample-induced losses overshadow other contributions, and it can be rigorously quantified. By comparing the Q-factor with and without the sample, we can isolate the inductive loss term, R_{m} .

When the probe is loaded with a conductive sample, the Q-factor becomes

$$Q_1 = \frac{\omega_0 L}{R_{\text{eff}}} = \frac{\omega_0 L}{R_{\text{m}} + R_{\text{extra}}} \Rightarrow \frac{1}{Q_1} = \frac{R_{\text{m}} + R_{\text{extra}}}{\omega_0 L}, \quad (2.3)$$

where $R_{\text{extra}} = R_{\text{coil}} + R_{\text{dielectric}} + R_{\text{radiative}}$ represents the probe's intrinsic losses.

Without the sample, the unloaded Q-factor is simply

$$Q_{\text{u}} = \frac{\omega_0 L}{R_{\text{extra}}} \Rightarrow \frac{1}{Q_{\text{u}}} = \frac{R_{\text{extra}}}{\omega_0 L}. \quad (2.4)$$

Subtracting Eq. 2.3 from Eq. 2.4, we isolate the inductive losses R_{m} ;

$$\boxed{R_{\text{m}} = \omega_0 L \left(\frac{1}{Q_{\text{l}}} - \frac{1}{Q_{\text{u}}} \right)}. \quad (2.5)$$

This expression links R_{m} to measurable changes in Q-factor, offering a direct path to quantify the sample's inductive effects. This expression will be employed throughout the thesis for the experimental determination of inductive losses.

2.2 Analytical Expressions for Inductive Losses

Equation 2.5 provides an experimental means to determine inductive losses. But how does conductivity factor in? These losses, driven by eddy currents, scale directly with the sample's electrical conductivity, σ , and we can express them analytically. To develop a quantitative understanding, we need a framework that relates the induced electromagnetic response to the coil's current distribution. A fundamental principle that aids in this analysis is the principle of reciprocity [15]. Reciprocity provides a mathematical shortcut: instead of directly computing the complex response of the sample to the coil, we can exploit a symmetry – relating the electromotive force (EMF) induced in a receiving coil by a dipole source at a specific point in space to the magnetic field that would be produced at that point by a unit current flowing through the coil (see Fig. 2.2). Since the latter is often easier to compute, reciprocity simplifies the derivation of inductive losses.

Existing literature often presents only partial derivations or final results, leav-

ing gaps in the theoretical development. I have re-derived the inductive loss for a spherical sample within a solenoid, as originally discussed in [5], and extended the analysis to cylindrical samples. A similar derivation has been conducted for a surface coil RF probe with a planar conductive sample. These derivations are detailed in Appendix A. These derivations serve as a first-principles approach to developing a deeper understanding of the underlying interplay between inductive loss, sample conductivity, and probe geometry. These derivations were the core of my summer 2024 work, prior to this honours project.

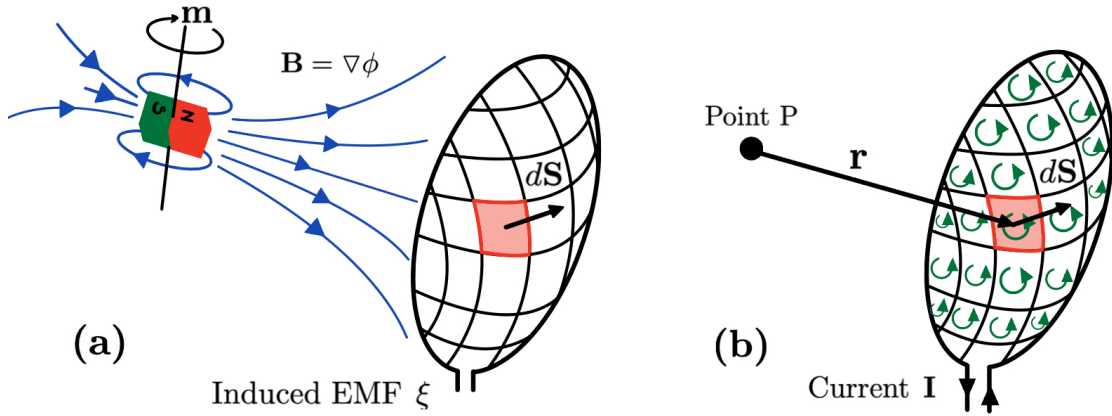


Figure 2.2: The principle of reciprocity in action: a closed surface edged by a conductor is split into a mesh. Each tiny element, with area dS (a vector normal to its plane), shrinks to zero in integrals. In (a), this mesh trick calculates the flux linkage from a nuclear magnetic moment $\mathbf{m}$ at point P. In (b), a current I looping the edge breaks into a mesh of circulating currents, where each element's magnetic moment $I dS$ creates a magnetic field – especially at point P (inspired by [13, 15]).

2.2.1 Solenoidal RF Probe

In MRI, Hoult and Lauterbur [5] applied the reciprocity principle to calculate the electrical noise generated by a lossy sample within a receiving coil. They modeled the system as a conducting sphere within a coil with uniform sensitivity B , assuming no dielectric losses. This assumption allowed them to analyze the problem by determining the power dissipation caused by a unit of alternating current at the coil's resonant frequency. Through this approach, they identified inductive losses

introduced by the sample into the probe circuit. Their results showed that for a conductive sphere (neglecting dielectric losses) inside a coil with uniform B , the losses scaled with the fifth power of the sphere's radius.

For a uniformly conducting spherical sample inside a solenoidal RF probe, as illustrated in Fig. 2.3a, the inductive loss is given by

$$R_m = \frac{\pi \omega_0^2 \mu_0^2 n^2 b^5 \sigma}{30 (a^2 + g^2)}, \quad (2.6)$$

where a and g are the coil's radius and half-length, respectively, b is the sphere's radius, n is the number of coil turns, and μ_0 is the permeability of free space. The final equation 2.6 is identical to the expression derived by Hoult and Lauterbur [5].

We assume a homogeneous field, ensuring uniformity across the sample volume. This simplification is reasonable for tightly wound solenoidal probes where field variations are minimal near the coil center. Additionally, the skin effect is ignored, as the skin depth at RF frequencies is significantly larger than the characteristic dimensions of the sample across a wide range of conductivities. This ensures that induced currents span the entire sample volume rather than being confined to the surface, making this approximation reasonable for the materials studied.

Similarly, for a conducting cylindrical sample within a solenoidal probe (Fig. 2.3b), under the same simplifying assumptions, the inductive loss is

$$R_m = \frac{\pi \omega_0^2 \mu_0^2 n^2 b^4 g \sigma}{16 (a^2 + g^2)}, \quad (2.7)$$

where b is now the cylinder's radius, and other variables retain their prior definitions. This result aligns with findings by Gadian and Robinson [16] and Cho [17], with Cho's formulation reducing to Eq. 2.7 despite differing variable conventions.

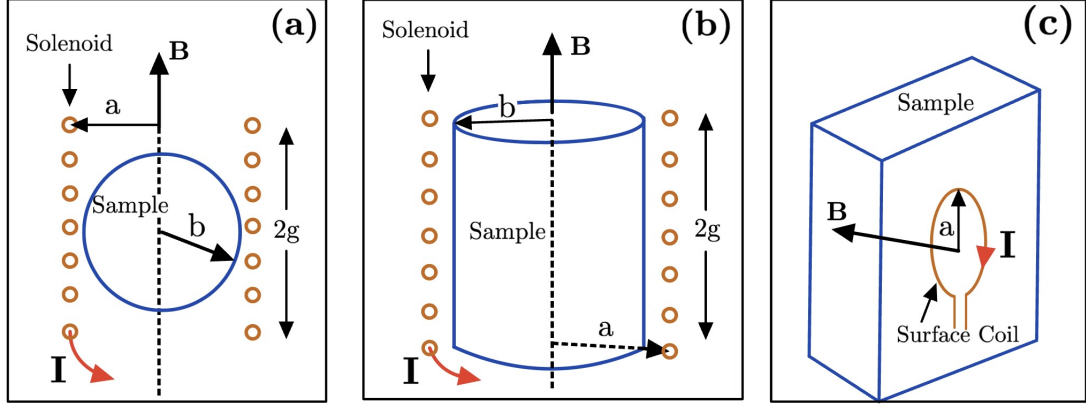


Figure 2.3: Schematic illustrations of conductive samples placed within RF probes, used for analytical calculations of inductive losses. (a) Spherical sample of radius b positioned centrally inside a solenoidal coil of radius a and length $2g$. (b) Cylindrical sample of radius b and length $2g$ placed centrally within the same solenoidal coil. (c) Planar conductive sample positioned adjacent to a surface coil of radius a (after [5, 11, 15]).

2.2.2 Surface Coil RF Probe

The simplest RF coil designs are circular loops known as surface coils, which generate a magnetic field perpendicular to the coil plane, with field amplitude decreasing along the coil axis. In contrast to solenoidal coils that produce approximately uniform fields ideal for volumetric samples, surface coils offer localized sensitivity tailored for planar or surface-constrained samples. Surface coils can also be miniaturized, providing versatility to accommodate samples of varying dimensions, especially when the sample geometry restricts the use of solenoids.

However, the strongly non-uniform magnetic fields produced by surface coils prevent the straightforward analytical approach used previously. Hayes and Axel [18] explored noise characteristics specifically in flat surface coils. They conducted experiments with circular surface coils placed either in contact with or near a flat tank filled with a conductive saltwater solution, empirically measuring noise as a function of coil radius. They proposed a functional form for sample resistance similar to that for solenoids:

$$R_m = \sigma \omega_0^2 f(a), \quad (2.8)$$

where σ represents the sample conductivity, ω_0 is the operating frequency, and $f(a)$ is an empirically determined function of coil radius a .

Extending the analysis from our earlier solenoidal coil study and addressing the challenges posed by the non-uniform field of surface coils, we use the magnetic vector potential approach to analytically derive the inductive loss caused by a planar conductive sample. The sample is assumed to be infinite laterally and in depth relative to the coil dimensions (see Fig. 2.3c). This simplifying assumption ensures maximal coupling between the coil's magnetic field and the conductive sample. In practice, if the sample size is insufficiently large, some magnetic field lines become stray fields extending beyond the sample boundaries, resulting in reduced coupling and an underestimate of the inductive losses and thus the measured conductivity. The derivation involves surface integrals over the coil's area to rigorously account for the non-uniform field distribution. The resulting analytical expression for inductive loss is

$$R_m = \frac{1}{3} \omega_0^2 \mu_0^2 n^2 a^3 \sigma . \quad (2.9)$$

where n is the number of turns of the coil and a is the radius of the surface coil. Harpen [19] approached the problem similarly, but employed a summation approximation for the integral calculation of the magnetic vector potential $\mathbf{A}$, using spherical harmonics expansions. His approximation yielded a slightly different inductive loss,

$$R_m = 0.328 \pi^2 \omega_0^2 \mu_0^2 n^2 a^3 \sigma, \quad (2.10)$$

differing from Eq. 2.9 by approximately 1%. Note that, unlike Eqs. 2.6 and 2.7, Eq. 2.9 is independent of sample size, making it suitable for samples large in depth and width relative to the surface coil size. We therefore aim to build smaller surface coil to make it versatile experimentally.

2.2.3 Implications of the Derived Analytical Expressions

The derived analytical expressions for inductive losses across all investigated configurations scale directly with the conductivity, σ , the square of the resonant frequency, ω_0 , and strongly depend on the characteristic sample radius, b or coil radius, a .

In subsequent experiments, we first select the desired operational frequency of the RF probe, generally aiming for higher frequencies to achieve higher measurement sensitivity. Given the chosen frequency, we then design and fabricate our probe to meet that target frequency. The probe geometry determines the inductance, and since the resonant frequency of the LC circuit (the RF probe) is:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (2.11)$$

[7, 8, 9, 14, 12], we adjust the probe geometry (thus changing the inductance, L) and select an appropriate capacitance, C to tune the probe to the desired frequency.

For a given probe design, examining Eqs. 2.6, 2.7, and 2.9 indicates that, aside from the conductivity σ of the sample, all other parameters are fixed by probe and sample geometry and the chosen frequency. Thus, the derived equations clearly suggest that inductive losses incurred in the sample should be directly proportional to its conductivity—a hypothesis we seek to validate experimentally. If this relationship holds true, it follows that we can reliably determine the conductivity of a sample simply by measuring the inductive losses it incurs.

Chapter 3

RF Probe Characterization

Let us now step back and visualize what happens to an RF probe's effective capacitance and inductance when a conductive sample is introduced. Recall that in our research, the RF probe plays a dual role: it transmits an oscillating magnetic field to induce eddy currents in the conductive sample and detects the resulting change in its resonance due to energy dissipation from those currents. All probes in this study are thus designed to transmit and detect these changes simultaneously.

3.1 Tuning and Matching the RF Probe

Consider an RF probe tuned to a frequency f_0 , where most (ideally all) of the RF signal from the system's power source is transmitted to the probe, with minimal reflection. This describes a perfectly tuned and matched probe. Tuning adjusts the probe's resonant frequency to match the signal frequency, optimizing sensitivity and accuracy. Matching aligns the probe's impedance with the system's, typically a pure resistive $50\ \Omega$, minimizing reflection and maximizing power transfer. A helpful analogy is picturing a plumber joining water pipes in a system: matching their diameters ensures smooth flow with minimal turbulence, much like aligning an RF probe's impedance for efficient signal transfer. Tuning, meanwhile, is like positioning

those pipes to avoid obstacles and optimize water flow—akin to adjusting the probe’s resonance to hit the right frequency. By analogy, the total losses of the probe and sample resemble internal friction in the pipes, which slows the water flow despite a perfect fit.

Now, place a conductive water-based sample (e.g., NaCl solution) near this tuned and matched probe. As discussed in § 2.1, the probe’s oscillating magnetic field induces eddy currents in the sample, which generate a secondary magnetic field opposing the primary field, per Lenz’s law [20]. This opposition reduces the effective inductance slightly, as the secondary field weakens the net magnetic flux through the coil [21]. Meanwhile, the sample – rich in charged ions and with a high dielectric constant (e.g., $\epsilon_r \approx 78.304$ at 25 °C for water [22]) – enhances capacitive coupling, increasing the effective capacitance.

To quantify this, RF engineers use indirect methods based on the reflection coefficient Γ :

$$\Gamma = \frac{Z - Z_0}{Z + Z_0}, \quad (3.1)$$

where $Z = R + iX$ is the probe’s complex impedance (with R as resistance and X as reactance), and $Z_0 = 50 \Omega$ is the system’s characteristic impedance, standard in RF systems like coaxial lines [12]. Clearly, $0 \leq |\Gamma| \leq 1$, where $|\Gamma| = 0$ corresponds to complete signal transmission (no reflection), and $|\Gamma| = 1$ represents total reflection with no transmitted power. The reflection coefficient versus frequency plot reveals these changes (Fig. 3.1a).

Returning to our thought experiment, these shifts in effective inductance and capacitance from the conductive sample disrupt the probe’s resonance. The resonant frequency may rise if inductance reduction prevails or fall if capacitance increase does (Eq. 2.11) – our tests showed a minor downward shift with NaCl solutions. More significantly, eddy currents raise the effective resistance (R_{eff}) through ohmic losses,

broadening the resonance curve and shifting the impedance away from $50\ \Omega$ [20]. This broadening, seen as a lower Q-factor, and the resulting mismatch – often toward over-coupling – reflect the sample’s energy dissipation, requiring retuning and rematching to restore optimal conditions [21].

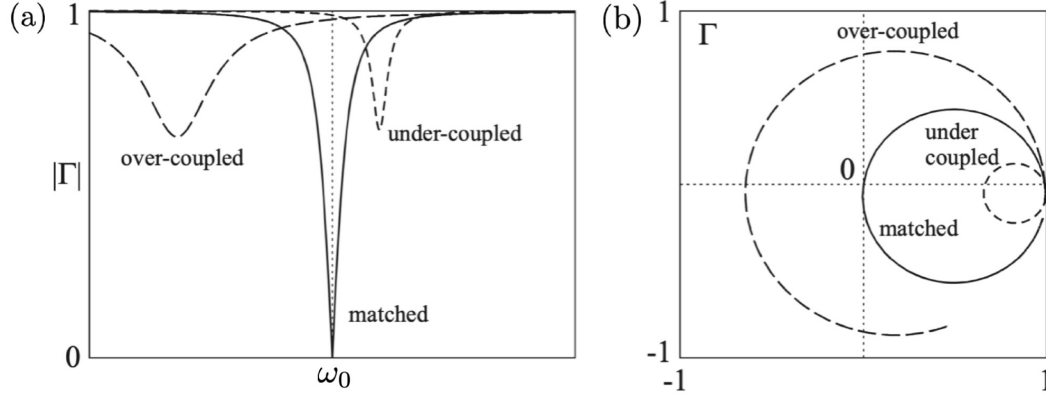


Figure 3.1: Reflection coefficient behavior under different coupling conditions. (a) The magnitude of the reflection coefficient, $|\Gamma|$, as a function of frequency for three coupling conditions: tuned and matched (solid line), under-coupled (dashed line), and over-coupled (dotted line). A well-matched probe exhibits minimal reflection at resonance (ω_0), while under-coupling and over-coupling result in higher reflections. (b) The same conditions are illustrated on a Smith chart, where the reflection coefficient’s position indicates the degree of impedance matching. The center of the chart corresponds to a perfect match ($\Gamma = 0$), while deviations indicate mismatches due to under- or over-coupling. In our case, introducing a conductive sample may shift the system toward the over-coupled regime (too much energy is coupled into the load), necessitating retuning and rematching to restore optimal conditions [12].

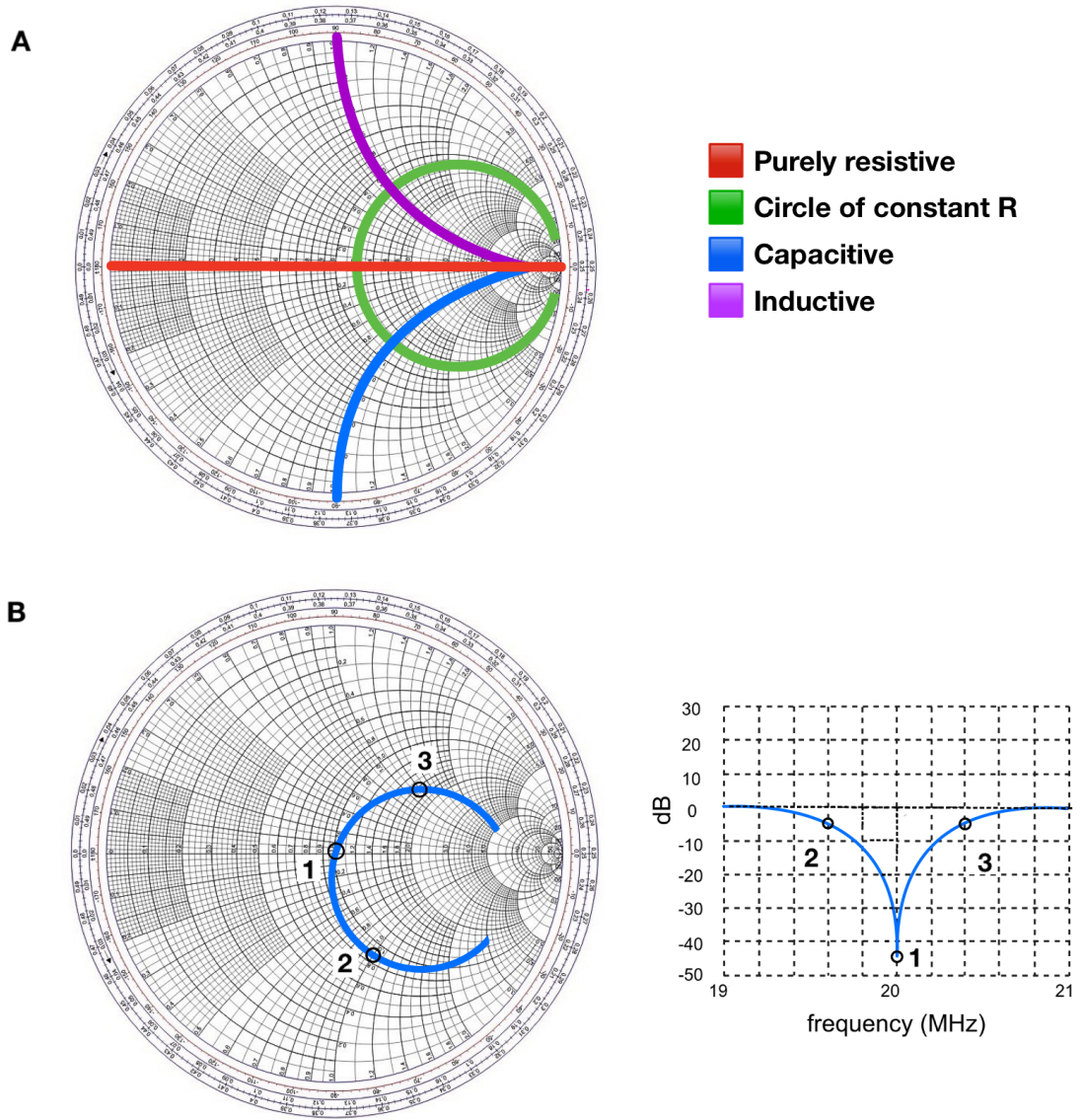


Figure 3.2: The Smith chart. (a) A general template (sourced from [scanalyst](#)) showing constant resistance circles (green) and reactance arcs (purple for inductive, blue for capacitive). The horizontal axis represents pure resistance, with the rightmost point indicating infinite impedance. (b) An example for a single-channel probe circuit tuned to 20 MHz. The right plot shows the reflection coefficient (Γ) versus frequency, with better matching (more negative Γ) at resonance. The left plot maps the impedance (blue) on the Smith chart, where marker 1 at resonance is purely resistive (center, $Z = 1$, normalized to $50\ \Omega$), and markers 2 and 3 show reactance shifts during tuning [7].

3.1.1 Retuning and Rematching the RF Probe

Continuing our visualization at the RF workbench, the RF probe – now detuned and mismatched by the conductive sample – awaits adjustment. Retuning and rematching begin by adjusting the matching capacitor C_M to minimize the reflection coefficient in the frequency response (Fig. 3.1a). In RF engineering, Γ is often expressed in decibels:

$$\Gamma_{\text{dB}} = 20 \log_{10}(|\Gamma|), \quad (3.2)$$

where a value below -20 dB (less than 1% reflected power) indicates a well-matched probe. The goal is to achieve this minimum dip, ensuring efficient signal transfer.

With a satisfactory match achieved, the tuning capacitor C_T is adjusted to shift the frequency to the desired value. If the matched frequency lies above the target f_0 , the tuning capacitance is increased; if below, it is decreased (per Eq. 2.11). This iterative process—adjusting C_M for matching and C_T for tuning—continues until the probe is perfectly tuned and matched, typically converging after a few cycles.

For precise matching below 1% reflection, the Smith chart provides a visual guide. It maps normalized impedance ($Z = Z_{\text{load}}/Z_0$), with the center ($\Gamma = 0$, $Z = 1$, or 50Ω) as the perfect match. Constant resistance circles and reactance arcs show deviations—capacitive below the horizontal axis, inductive above (Fig. 3.2). Introducing the conductive sample shifts the impedance, often toward the over-coupled regime, as seen in Fig. 3.1b. Using the Smith chart, the matching capacitor is adjusted to guide the impedance toward the center, minimizing reflection, while the tuning capacitor aligns the resonance with f_0 . In practice, these two techniques—frequency response minimization and Smith chart analysis—are combined to efficiently retune and rematch the probe. Figure 3.3 shows the frequency response, comparing the initially tuned and matched RF probe to its retuned and rematched state after sample introduction.

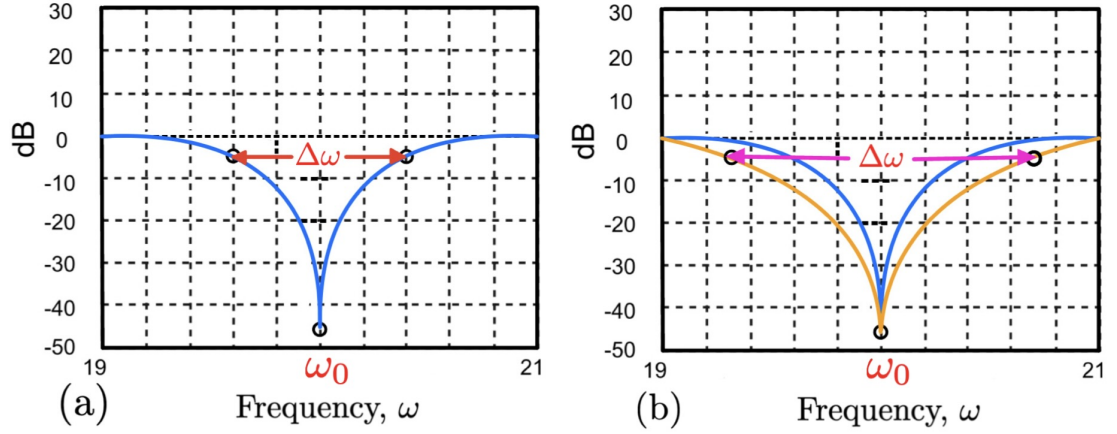


Figure 3.3: Frequency response of a tuned and matched RF probe before and after loading with a conductive sample. (a) The initial response (blue) shows a well-tuned and well-matched resonance at ω_0 , with a sharp dip in the reflection coefficient (Γ_{dB}) and a narrow -3 dB bandwidth $\Delta\omega$, where the power drops to half its peak value. (b) After loading the probe with a conductive sample and performing retuning and rematching (orange), the resonance shifts and the bandwidth $\Delta\omega$ widens due to increased losses in the sample.

3.2 Experimental Measurement of Q-Factor

Building on the tuned and matched RF probe, let's explore how its Q-factor is measured. The filling factor η , defined as the ratio of the Q-factor with a sample loaded (Q_1) to that when empty (Q_u), quantifies the probe's coupling with the sample. The probe's sensitivity is approximately proportional to $\sqrt{\eta Q}$, making Q-factor measurement a key step in evaluating probe sensitivity in MR [6, 12, 15]. The filling factor will be used in subsequent experiments using solenoidal RF probe to optimize the probe's sensitivity, ideally approaching unity when the sample fully occupies the coil's sensitive volume. However, we use a surface coil to escape the issue of filling factor.

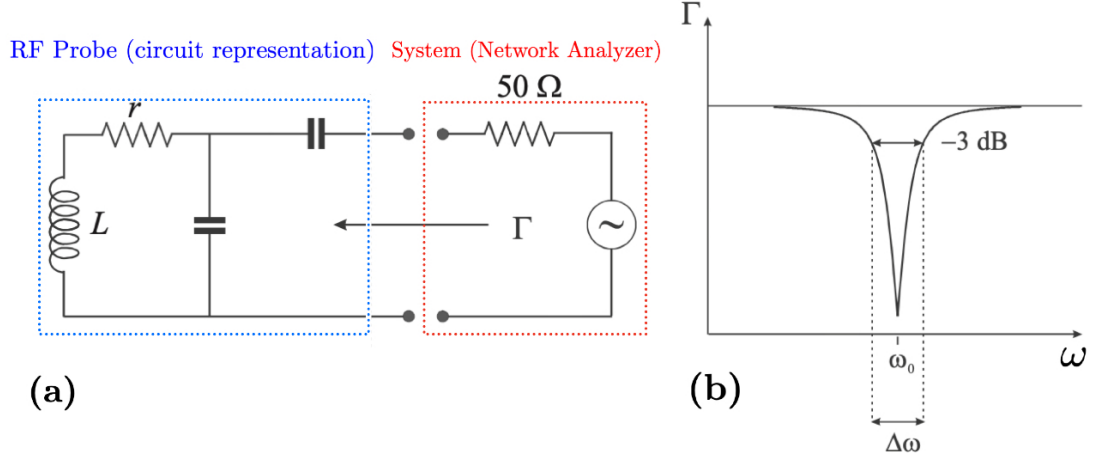


Figure 3.4: Schematic of a matched RF probe resonator and its reflection coefficient frequency response. The left panel (a) depicts the circuit diagram of a matched probe resonator used in this study, featuring inductance (L), resistance (r), and capacitors tuned to a $50\ \Omega$ impedance. The right panel (b) shows the reflection coefficient (Γ) frequency response, with the bandwidth ($\Delta\omega$) at the -3 dB point – where power drops to half its peak – used to calculate the Q-factor (after [12]).

While several methods exist to determine the Q value, it is often directly evaluated on the RF workbench from the frequency response of the reflection coefficient, $\Gamma_{\text{dB}}(\omega)$, at the -3 dB level-half-maximum power level (Fig. 3.4b). This measurement is done when the probe is matched to the characteristic impedance, Z_0 ($50\ \Omega$). In such cases, the actual Q value of the probe resonator (which is related to losses in the inductor resistance) is double the Q measured at the -3 dB level. This is because the matched resonator is damped by the characteristic load impedance Z_0 connected to the circuit. Consequently, half of the energy is dissipated in the source load, and the other half in the matched impedance of the probe resonator. Therefore,

$$Q = \frac{\omega_0 L}{R_{\text{eff}}} = \frac{2\omega_0}{\Delta\omega}. \quad (3.3)$$

Thus, measuring the Q value involves determining the resonant frequency and the bandwidth of the RF probe when it is interacting with the sample. Typically, the equation $Q = \omega_0/\Delta\omega$ is used when focusing solely on the intrinsic properties of the probe and sample, without considering impedance matching to the measurement

system. However, this ignores reflection losses. Therefore, Eq. 3.3 is used since it accounts for the damping effects of the characteristic load impedance, ensuring accurate and comprehensive evaluation of the probe's performance in a matched system. With the groundwork in RF probe characterization established, we now shift to the core of the investigation – testing these theoretical and conceptual foundations through experiments.

Chapter 4

Experiments and Results

The ultimate test of any physical theory is, of course, experimentation; successful validation opens the path to practical applications (after [23]). This chapter, therefore, presents a series of basic experiments to evaluate our conductivity measurement. We begin by constructing two simple RF probes, a solenoidal and a surface coil probe, and preparing salt solution samples with varying conductivity to verify our model equations. Next, we repeat the experiments using the same surface coil probe with a CaCl_2 solution to ensure inductive loss do not depend on ion type. They should ideally depend solely on conductivity. We then explore a loop-gap resonator, tunable to higher frequencies than previous designs, and compare its performance with the earlier solenoidal probe. Finally, we conducted a few preliminary experiments to assess potential practical applications of this conductivity measurement technique.

In most experiments, an RF Network Analyzer (300 kHz to 3000 MHz) model 8714B (Hewlett Packard, Santa Clara, CA) was used to assist with tuning, matching, and measuring the circuit's Q-factor and an LC Meter model MLC500 (Dongguan Jingyan Instruments & Technology Co., Ltd., Kwai Chung, N.T.) was employed to measure the inductance of the RF coil. A detailed discussion on the propagation of uncertainty in inductive losses is presented in Appendix B.

4.1 Initial Tests

The initial tests explored whether the measured inductive losses aligned with the predicted linear dependence on conductivity. Note that the probe design and experimental discussions are thoroughly detailed in the first experiment. Subsequent experiments adopt the same design and a comparable approach, and thus are not elaborated in detail.

4.1.1 Solenoidal RF Probe with NaCl Solution Sample

A ten-turn solenoid, 50 mm in length and 50.3 mm in diameter, wound from copper wire, was tuned and matched to 10 MHz using variable capacitors and a 5 pF fixed high-Q air-spaced capacitor on the tuning side. The coil’s inductance, measured with an LC meter, was $3.42 \pm 0.04 \mu\text{H}$. This value, taken without the sample, serves as our baseline for calculating inductive losses. Introducing a conductive sample slightly reduces the effective inductance due to eddy currents, but this reduction—typically 5-10% in RF coils [20]—falls within our measurement uncertainty and minimally impacts R_m . This approach aligns with standard RF resonator analysis where inductance variations are small compared to resistive effects [21].

A cylindrical sample tube (outer diameter 40 mm, length 64.5 mm) was placed within the solenoid, and the circuit’s Q value was measured for samples with varying sodium chloride concentrations, including distilled water. A salinity range of 1–4% was selected. For perspective, freshwater sources such as rivers and lakes typically contain less than 0.05% salt, while seawater has an average salinity of approximately 3.5% [24, 25]. The RF circuit was housed in a separate tuning box, which served as a Faraday cage to prevent electric field lines, associated with distributed capacitance, from passing through the sample, thereby eliminating dielectric losses. However, its proximity to the RF coil may introduce a small additional loss.

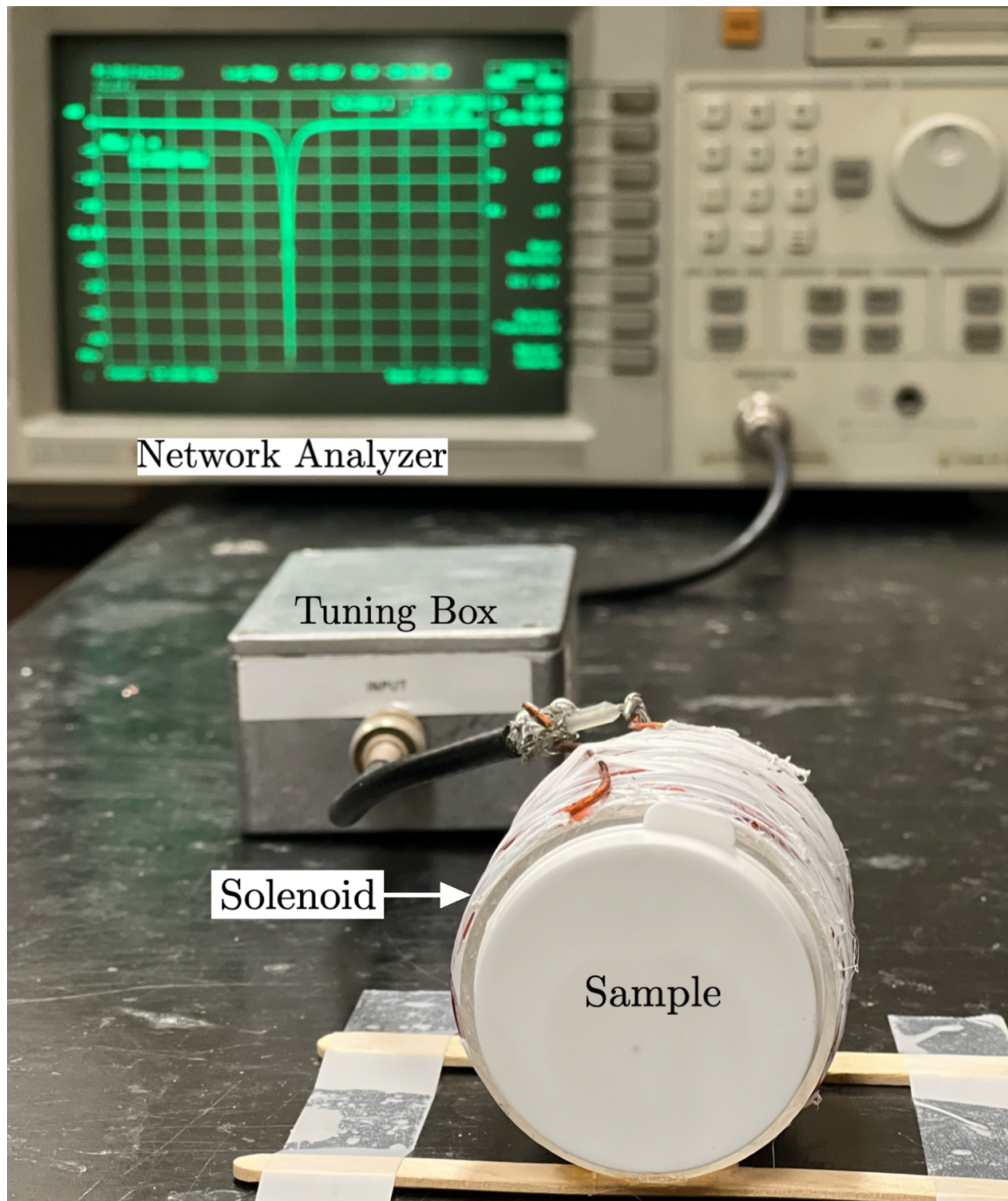


Figure 4.1: Experimental setup of a solenoidal RF probe tuned to 10 MHz, with the sample placed inside the solenoid. The solenoid is connected to a tuning box and a network analyzer for Q-factor measurement, conducted on a laboratory bench.

During my 2024 summer research, I measured the Q -factor of samples at varying heights within a solenoidal probe. This approach revealed that positioning the sample at the probe’s center represents the optimal “sweet spot”, where electromagnetic interaction is maximized [26]. This maximum interaction, indicated by increased bandwidth broadening (maximal loss), significantly reduces the Q value. We therefore sought sample placement around the probe’s center. Additionally, as discussed in § 3.2, enhanced sensitivity is ideally observed when the sample fills the probe, a condition reflected in our choice of sample and probe sizes. The experimental setup is shown in Figure 4.1.

Results and Discussions

Table 4.1 demonstrates the influence of increasing NaCl concentration on probe loading. As salinity increases, the Q -factor decreases due to increased conductive losses, which directly correspond to an increase in inductive loss R_m . The trend follows the expected behavior of a lossy conductive medium interacting with an RF probe, where increased salinity enhances energy dissipation. The uncertainties, primarily influenced by Q -factor measurement variability and inductance precision, contribute to about $0.02 \, \Omega$ uncertainty in R_m , yet the overall trend remains consistent with theoretical expectations for inductive losses in conductive samples. The expected uncertainties stem from analyzing the measurement process, not from conducting repeated measurements.

Table 4.2, on the other hand, compares the total effective resistance, $R_{\text{eff}} = \frac{\omega_0 L}{Q}$ with the inductive losses R_m , obtained from Eq. 2.5, for NaCl solutions fully inserted into the RF coil. The table also includes theoretical inductive losses, $R_{m,\text{theory}}$ calculated using Eq. 2.7, which assumes uniform sample placement (i.e., equal distribution on both sides of the probe). However, as shown in Fig. 4.1, the sample extends beyond the probe due to the cap, introducing asymmetry. This suggests that the theoretical values are approximate and may not fully reflect the experimental setup.

Table 4.1: Measured frequency shifts (Δf), Q-factors, inverse Q-factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a solenoidal RF probe with $L = 3.42 \pm 0.04 \mu\text{H}$ and $f_0 = 10.000 \pm 0.001 \text{ MHz}$.

NaCl Density (%)	Δf ($\pm 0.001 \text{ MHz}$)	Q-factor	$\Delta \frac{1}{Q}$ (± 0.0001)	R_m (Ω)
Empty Probe	0.098	204 ± 3	...	...
0	0.100	200 ± 3	0.0001	0.02 ± 0.02
1	0.140	143 ± 1	0.0021	0.45 ± 0.02
2	0.171	117 ± 1	0.0037	0.78 ± 0.02
3	0.197	102 ± 1	0.0050	1.06 ± 0.02
4	0.228	88 ± 1	0.0065	1.40 ± 0.03

Note: $\Delta \frac{1}{Q} = \frac{1}{Q_1} - \frac{1}{Q_u}$ and $R_m = \omega_0 L \cdot \Delta \frac{1}{Q}$

Table 4.2: Comparison of the total effective resistance R_{eff} with inductive losses R_m , obtained from Eq. 2.5, for NaCl solutions fully inserted into the RF coil. The constant difference $R_{\text{extra}} = R_{\text{eff}} - R_m$ supports the assumption that R_m arises solely from the sample. The table includes theoretical inductive losses ($R_{m,\text{theory}}$) from Eq. 2.7, assuming uniform sample placement. The experimental uncertainty in R_m is approximately $\pm 0.02 \Omega$.

NaCl Density (%)	Experimental			$R_{m,\text{theory}}$ (Ω)
	R_{eff} (Ω)	R_{extra} (Ω)	R_m (Ω)	
0	1.13	1.11	0.02	0.00
1	1.58	1.13	0.45	0.45
2	1.93	1.15	0.78	0.76
3	2.23	1.16	1.06	1.08
4	2.58	1.18	1.40	1.40

Fig. 4.2 presents the initial analysis of inductive losses R_m as a function of NaCl density (in mass percent). The linear trend observed indicates that inductive loss increases with salinity, consistent with the known proportionality between NaCl density and conductivity. This relationship underpins the subsequent conductivity-based analysis in Fig. 4.3, where the theoretical model (Eq. 2.5) is further verified against experimental data.

The conductivity at each salinity level was determined from the slope and intercept of a conductivity-versus-salinity calibration curve for NaCl solutions [27]. However, the high conductivity value (y-intercept) at 0% salinity, based on this method, requires scrutiny, as the data in [27] begins at 0.5% salinity. To address this, we adopted the distilled water conductivity value of $0.04193 \mu\text{S}/\text{cm}$ from [28]. All literature conductivity values are referenced at 20°C .

Fig. 4.3 illustrates the relationship between inductive loss, R_m and conductivity, σ of NaCl solutions. The black data points represent experimental measurements of inductive losses, while the red line shows the theoretical fit from Eq. 2.5, which predicts zero inductive loss at zero conductivity. The least-squares fit yields a slope of $0.240 \pm 0.003 \Omega \text{ m}/\text{S}$, reflecting the probe's sensitivity to conductivity changes, with an intercept of $0.02 \pm 0.01 \Omega$. The fit's non-zero intercept results in a small overestimation of conductivity at 0% salinity, likely due to the sample container's contribution plus minor systematic offsets. The standard deviation of the residuals is 0.02Ω , comparable to the measurement uncertainty (i.e., $0.02\text{-}0.03 \Omega$), indicating a reasonable fit despite potential asymmetries in sample placement. This agreement validates Eq. 2.5 for estimating inductive losses under similar conditions.

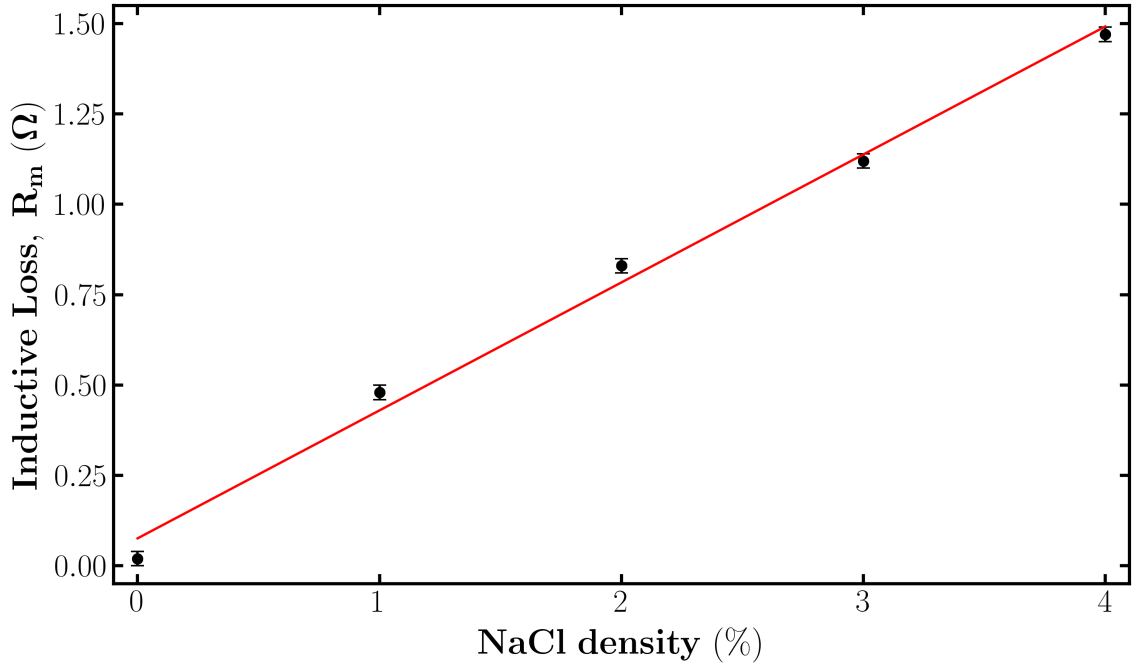


Figure 4.2: Inductive losses (R_m) measured for NaCl solutions, plotted against NaCl density (%). The least squares fit (red) models the linear increase in inductive loss with salinity.

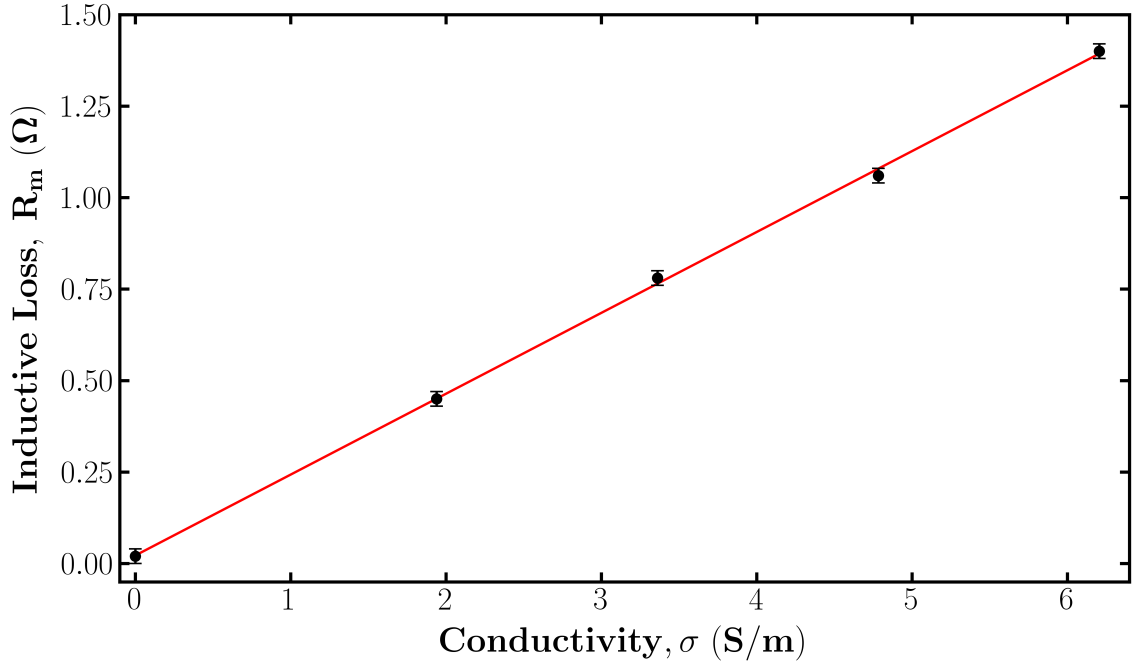


Figure 4.3: Inductive losses measured for NaCl solutions, plotted against conductivity. The fitted curve (red), derived from Eq. 2.5, has an intercept of $0.02 \pm 0.01 \Omega$, indicating a strong linear relationship between conductivity and inductive loss, with a slope of $0.240 \pm 0.003 \Omega \text{ m/S}$.

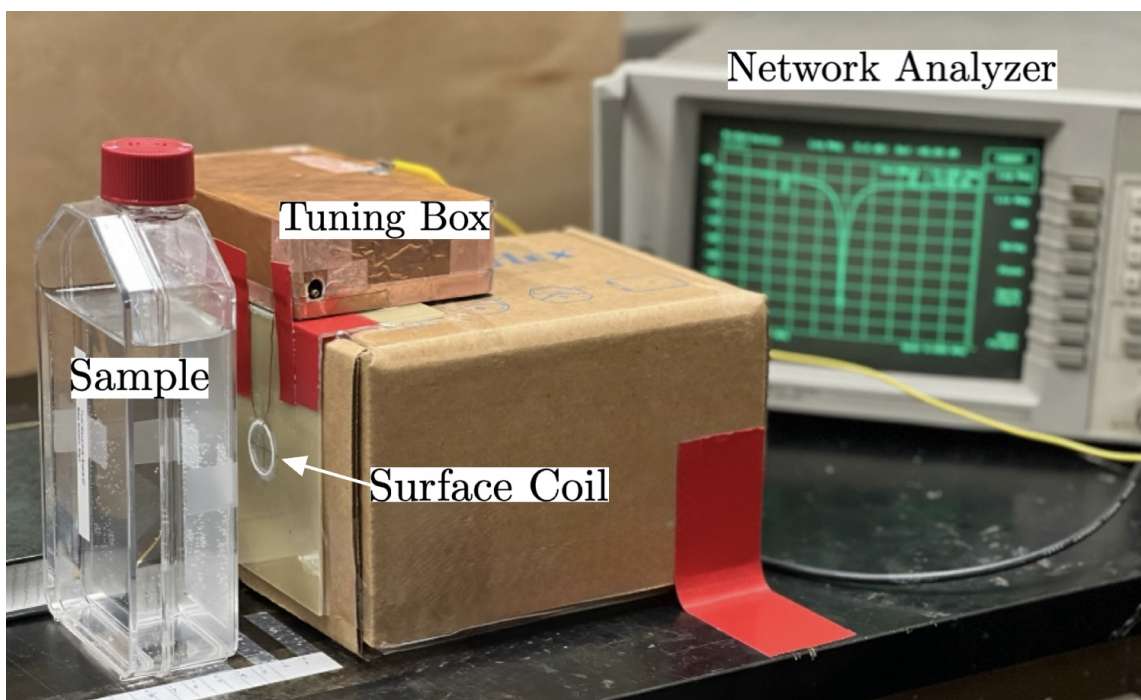


Figure 4.4: Photograph of the experimental setup, showing the effectively infinite planar sample positioned near the surface coil RF probe tuned at 50 MHz. Note: In the actual experiment, the sample enclosure is in direct contact with the surface coil.

4.1.2 Surface Coil RF Probe with NaCl Solution Sample

A two-turn surface coil with a 1 cm radius and an inductance of $0.344 \pm 0.002 \mu\text{H}$ was constructed from copper wire, tuned to 50 MHz, and matched to 50Ω using variable capacitors. Fixed capacitors of 15 pF and 20 pF were incorporated on the series matching and parallel tuning sides, respectively. Five NaCl solution samples of varying salinities, including distilled water, were prepared in a 600 ml cell culture flask with approximate dimensions of $10.5 \times 3 \times 14 \text{ cm}$. The coil was mounted in contact with the center of one vertical side of the flask, oriented such that the normal to the coil's plane was perpendicular to the flask's surface, aligning the time-varying magnetic field perpendicular to the surface to maximize field penetration into the sample (see Fig. 4.4). The Q-factor of the circuit was measured for each sample.

Results and Discussion

Table 4.3 presents the measured frequency shifts (Δf), Q-factors, inverse Q-factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity, consistent with the approach outlined in § 4.1.1. Following the procedure in § 4.1.1, we converted salinity to conductivity. Figure 4.5 shows a strong correlation between inductive losses, R_m and conductivity, σ for NaCl solutions up to 4% salinity. This is reinforced by the excellent fit of Eq. 2.9 to experimental data, confirming $R_m \propto \sigma$ with a slope of $0.30 \pm 0.01 \Omega\text{mS}^{-1}$ and an intercept of $0.070 \pm 0.004 \Omega$. Note that theoretically, $R_m \propto \omega_0^2 \sigma \int_V B^2 dV$, where $\int_V B^2 dV$ represents the integral of the squared magnetic field over the sample volume ([5, 12]).

Compared to the 10 MHz solenoidal probe (slope $0.240 \pm 0.003 \Omega\text{mS}^{-1}$, § 4.1.1), the surface coil at 50 MHz shows a 25% increase in slope, consistent with the expected frequency scaling ($(50/10)^2 = 25$), though moderated by differences in $\int_V B^2 dV$ due to the surface coil's less uniform field distribution (see Table 4.6 for comparison). The standard deviation of the residuals between the measured data and the fitted curve is 0.04Ω , indicating reasonable model accuracy despite a slight non-zero intercept (approximately 0.07Ω). This intercept, also observed at 0% salinity, is likely due to the distilled water's minor conductivity ($\sim 10^{-6} \text{ S/m}$) from impurities or dissolved CO_2 , which is not surprising given typical water purity levels. Otherwise, this agreement validates Eq. 2.9 for surface coil configurations, supporting its applicability in RF probe loading studies.

Table 4.3: Measured frequency shifts (Δf), Q-factors, inverse Q-factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a surface coil RF probe with $L = 0.344 \pm 0.002 \mu\text{H}$ and $f_0 = 50.000 \pm 0.001 \text{ MHz}$.

NaCl Density (%)	Δf ($\pm 0.001 \text{ MHz}$)	Q-factor	$\Delta \frac{1}{Q}$ (± 0.00002)	R_m (Ω)
Empty Probe	0.676	148.0 ± 0.3	...	...
0	0.738	135.5 ± 0.3	0.00060	0.067 ± 0.004
1	1.250	80.2 ± 0.1	0.00570	0.62 ± 0.04
2	1.686	59.30 ± 0.05	0.01010	1.1 ± 0.1
3	2.104	47.50 ± 0.03	0.01430	1.5 ± 0.1
4	2.453	40.80 ± 0.02	0.01780	1.9 ± 0.1

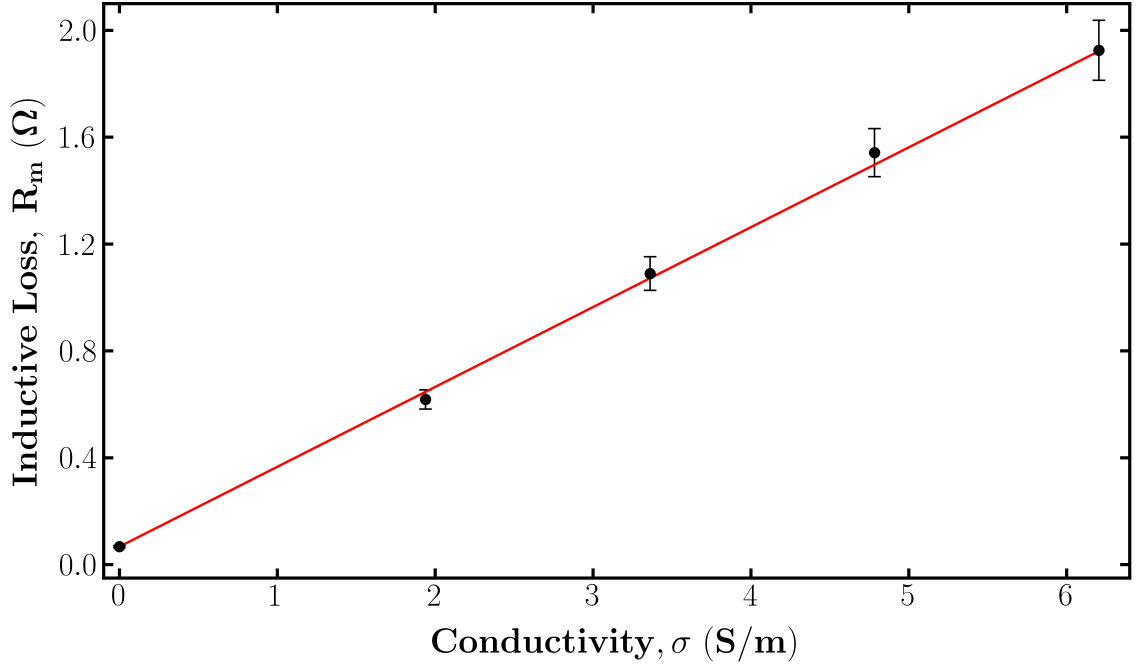


Figure 4.5: Inductive losses measured within NaCl solution samples using the circular surface coil RF probe at 50 MHz, plotted against conductivity. The fitted line (red), derived from Eq. 2.9, has an intercept of $0.070 \pm 0.004 \Omega$ and a slope of $0.30 \pm 0.01 \Omega \text{mS}^{-1}$.

4.1.3 Surface Coil RF Probe with CaCl₂ Solution Sample

In experiments described earlier, we showed the efficacy of a non-invasive, non-contact RF probe loading method for measuring the conductivity (§ 4.1.2). To extend this approach, we aim to confirm its applicability to other salt solutions. In this experiment, we conducted conductivity measurements for CaCl₂ solutions and compared the results with those of NaCl solutions. Since the RF signal losses R_m in the receiving coil are directly linked to the sample's conductivity σ ($R_m \propto \sigma$), rather than the specific type of ions present, we hypothesize that the results for CaCl₂ solutions will agree with those for NaCl solutions.

The same surface coil RF probe from § 4.1.2, tuned to 50 MHz, was used in this experiment. Five samples each of NaCl and CaCl₂ solutions, including distilled water, were prepared in 600 mL cell culture flasks with approximate dimensions of $10.5 \times 3 \times 14$ cm, consistent with the NaCl experiment. Sample conductivity was also measured with a DDS-307 desktop conductivity meter (Hefei Chenchuang Instrument Co., Ltd., Hefei), an electrode-based device. For each sample, the Q-factor of the RF probe was measured in both unloaded and loaded conditions.

Results and Discussion

Table 4.4 compares the conductivity values from literature [27, 28], desktop meter measurements, and the RF probe loading technique. The RF probe loading conductivity values were calculated using the model equation (Eq. 2.9):

$$R_m = \frac{1}{3}\omega_0^2\mu_0^2n^2a^3\sigma \quad \Rightarrow \quad \sigma = \frac{3R_m}{\omega_0^2\mu_0^2n^2a^3}, \quad (4.1)$$

where μ_0 is the permeability of free space, n is the number of turns of the coil, a is the radius of the surface coil, and σ is the conductivity of the sample. The close agreement between the desktop EC meter measured conductivity values and the literature values is no surprise. But the consistency between these values and those

obtained using our RF probe loading approach reinforces confidence in the method’s reliability.

Figure 4.6 presents the relationship between inductive losses, R_m and conductivity, σ for NaCl and CaCl₂ solutions. A least-squares fit to the combined data, using Eq. 2.9, yields a slope of $0.110 \pm 0.002 \text{ } \Omega \text{mS}^{-1}$, lower than the $0.30 \pm 0.01 \text{ } \Omega \text{mS}^{-1}$ from NaCl alone (§ 4.1.2). This decrease mainly arises from differences in sample containers, both positioned face-to-face with the surface coil [20]. The earlier experiment used a thin low-density polyethylene (LDPE) container, while this one used thicker polystyrene, causing the coil’s non-uniform magnetic field to decay further before coupling with the sample. Although the container faces touch the coil, the thicker wall increases the sample’s distance from the field’s peak, reducing R_m per unit conductivity. Additionally, the smaller sample volume here (450 mL vs. 600 mL previously) may play a role. While Eq. 2.9 assumes an infinite sample laterally and in depth, the larger 600 mL volume better approximates infinite lateral extent—despite comparable depths—enhancing field coupling and thus R_m in the earlier setup [21].

The standard deviation of the residuals between the measured data and the fitted curve is $0.02 \text{ } \Omega$, indicating reasonable model accuracy though a slight non-zero intercept ($= 0.070 \pm 0.002 \text{ } \Omega$) is observed here as well. The close alignment of R_m for both salts confirms our hypothesis that inductive losses depend primarily on conductivity, not ion type, supporting the RF probe loading method’s applicability across different ionic solutions.

Table 4.4: Comparison of conductivity values from literature [27, 28], DDS-307 electrode based desktop conductivity meter measurements, and our RF probe loading technique for NaCl and CaCl₂ solutions at various concentrations in mass percent.

Density (%)	Literature (S/m)		Desktop Meter (S/m)		Probe Loading (S/m)	
	NaCl	CaCl ₂	NaCl	CaCl ₂	NaCl	CaCl ₂
0	4.2×10^{-6}	4.2×10^{-6}	0.00012	0.00012	0.0 ± 0.1	0.0 ± 0.1
1	2.0	1.7	1.7	1.7	2.1 ± 0.1	1.7 ± 0.1
2	3.4	3.0	3.2	3.2	3.3 ± 0.2	3.4 ± 0.2
3	4.8	4.1	4.4	4.6	4.8 ± 0.3	4.4 ± 0.3
4	6.2	5.4	6.0	6.0	6.0 ± 0.3	5.2 ± 0.3

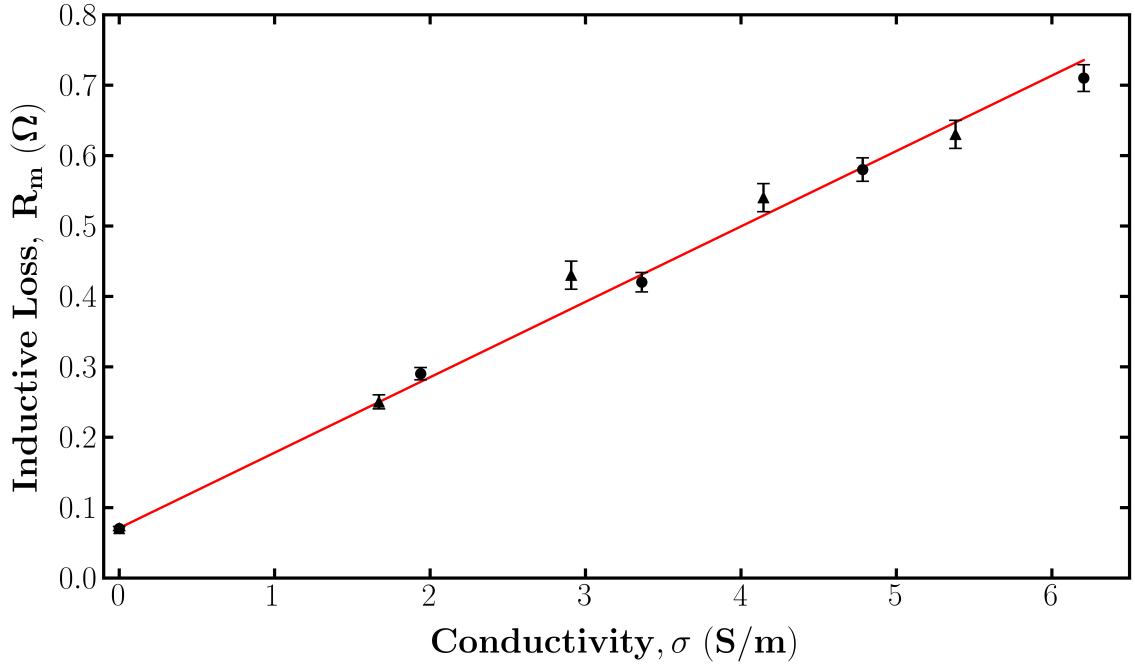


Figure 4.6: Inductive losses measured for CaCl₂ (▲) and NaCl (●) solutions using the 50 MHz surface coil RF probe, plotted against conductivity. The blue line represents the least-squares fit to the combined data, derived from Eq. 2.9, with a slope of $0.110 \pm 0.002 \text{ } \Omega\text{mS}^{-1}$ and an intercept of $0.100 \pm 0.002 \text{ } \Omega$.

4.1.4 Loop-Gap Resonator

As established in Eqs. 2.7 and 2.9, the inductive loss R_m is proportional to the square of the resonant frequency ($R_m \propto \omega_0^2$). This relationship suggests that increasing the probe’s frequency enhances its sensitivity to conductivity variations. While solenoidal RF probes provide uniform magnetic fields and strong coupling, increasing their resonant frequency requires reducing either inductance or capacitance (Eq. 2.11). However, reducing capacitance may limit tuning and matching, making inductance the primary parameter to adjust. Achieving a significant reduction in L while maintaining sample compatibility and field homogeneity is challenging.

A promising alternative is the Loop-Gap Resonator (LGR) [29, 30], which offers lower inductance while preserving B-field uniformity. LGRs function as single-turn solenoids with integrated capacitive gaps, naturally achieving high frequencies without excessive reductions in capacitance. They have been widely adopted in electron spin resonance (ESR) and high-frequency NMR due to their high Q-factors and efficient sample coupling [30, 31]. To assess the sensitivity enhancement in inductive loss measurements, we designed an LGR with the same dimensions as the 10 MHz solenoidal probe tested earlier (§ 4.1.1).

Results and Discussions

The LGR was constructed by wrapping a 0.025 mm thick, 50 mm $\times$ 157 mm copper foil around a 50 mm outer diameter (43.6 mm inner diameter), 60 mm long cylindrical plastic form. The gap between the two ends of the foil was approximately 0.4 mm. Hornak and others [29] designed similar LGRs using a single-turn copper loop mounted on a plastic sleeve, allowing for impedance matching by adjusting the loop’s distance from the resonant cylinder. In their approach, tuning capacitors were placed directly across the gap in the copper cylinder.

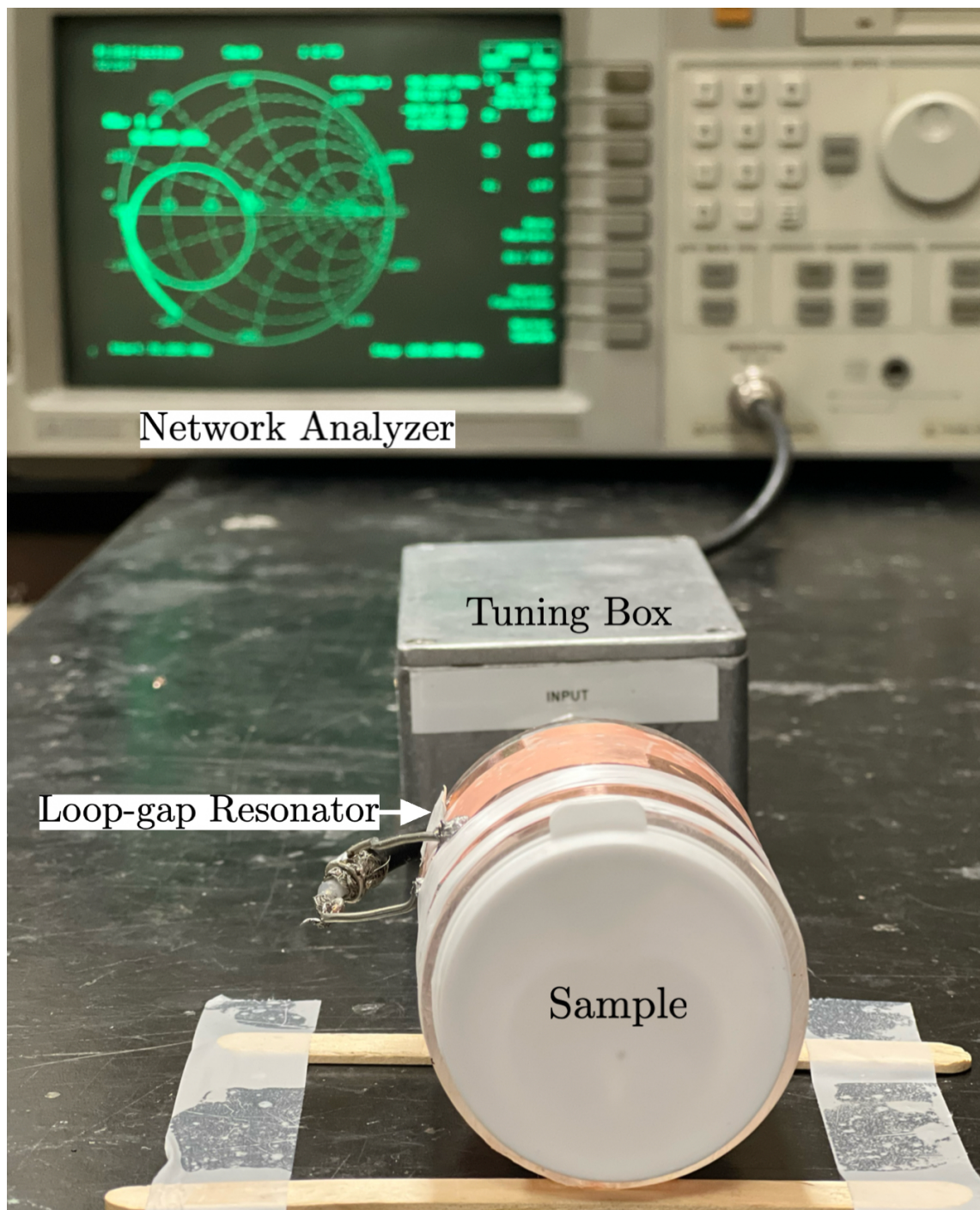


Figure 4.7: Photograph of the experimental setup, showing a cylindrical sample inside LGR tuned at 90 MHz.

Initially, I implemented this conventional method, but while tuning was successful, matching proved challenging, particularly when the probe was loaded with saline samples. To address this, I adapted the approach used in my earlier probes. Instead of relying on loop-based coupling, I directly connected the tuning box, which contained variable capacitors for both tuning and matching, to the two ends of the gap in the copper foil. This parallel-tuned, series-matched configuration was successful, with precise tuning to 90 MHz and stable impedance matching to $50\ \Omega$ across all tested conditions.

The measured data in Table 4.5 show a clear trend: as salinity increases, the Q-factor decreases while the inductive loss increases, consistent with expectations for conductive loading. Fig. 4.8 plots the inductive losses for the LGR probe against NaCl solution conductivity, showing a linear relationship with a slope $m = 0.20 \pm 0.01\ \Omega\text{mS}^{-1}$ and an intercept of $0.00 \pm 0.03\ \Omega$, derived from a least-squares fit to $R_m = m\sigma + c$ (with c being the intercept). In contrast, Fig. 4.3 presents the inductive losses for the 10 MHz solenoidal probe, with a slope of $0.240 \pm 0.003\ \Omega\text{mS}^{-1}$.

Contrary to expectations, the LGR's slope is slightly lower than the solenoid's, despite a frequency jump from 10 MHz to 90 MHz (see Table 4.6 for comparison). As discussed previously, $R_m \propto \omega_0^2 \sigma \int_V B^2 dV$ theoretically [5, 12], while experimentally, losses are determined from $R_m = \omega_0 L \left(\Delta \frac{1}{Q} \right)$. The frequency ratio, $\left(\frac{90}{10} \right)^2 = 81$, suggests a potential increase in slope. However, the LGR's single-turn design produces a weaker magnetic field distribution ($\int_V B^2 dV$) than the multi-turn solenoid's, as B scales with turn count [20], and its lower inductance ($0.085\ \mu\text{H}$ vs. $3.420\ \mu\text{H}$) further tempers R_m sensitivity to σ . This offsets the ω_0^2 gain, yielding the observed slope. While the LGR's 90 MHz operation ensures uniform field distribution, its reduced sensitivity reflects a trade-off between frequency and coupling strength, suggesting geometry optimization (e.g., increasing $\int_V B^2 dV$ through design adjustments) to better harness the frequency advantage [21].

Table 4.5: Measured frequency shifts (Δf), Q-factors, inverse Q-factor differences ($\Delta(1/Q)$), and inductive losses (R_m) for NaCl solutions of varying salinity (in mass percent), using a LGR RF probe with $L = 0.085 \pm 0.003 \mu\text{H}$ and $f_0 = 90.000 \pm 0.001 \text{ MHz}$.

NaCl Density (%)	Δf ($\pm 0.001 \text{ MHz}$)	Q-factor	$\Delta \frac{1}{Q}$ (± 0.00001)	R_m (Ω)
Empty Probe	2.081	86.5 ± 0.1	...	...
0	2.082	86.5 ± 0.1	0.00001	0.000 ± 0.001
1	3.180	56.60 ± 0.03	0.00610	0.30 ± 0.02
2	4.145	43.4 ± 0.01	0.01147	0.55 ± 0.03
3	5.150	35.00 ± 0.01	0.01705	0.82 ± 0.05
4	5.720	31.50 ± 0.01	0.02020	0.97 ± 0.06

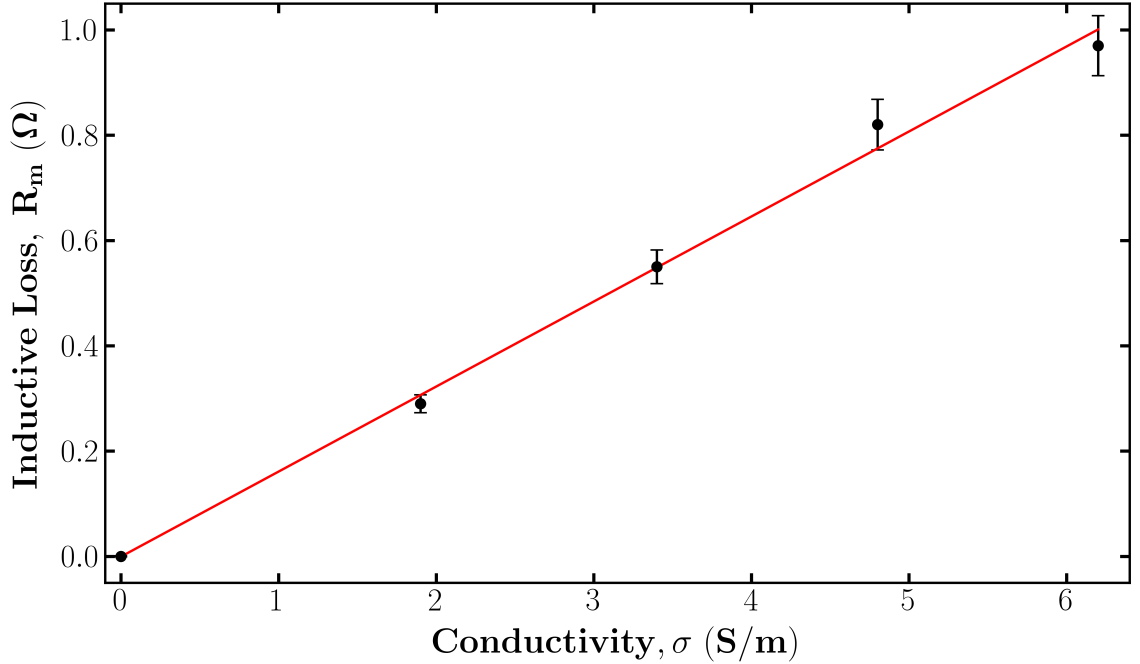


Figure 4.8: Inductive losses measured in NaCl solutions using the LGR RF probe, plotted against conductivity. The least-squares fit line (red) has a slope of $0.20 \pm 0.01 \Omega\text{mS}^{-1}$ and an intercept of $0.00 \pm 0.03 \Omega$, indicating a direct proportionality between conductivity and inductive loss.

Table 4.6: Comparison of slopes R_m/σ for different RF probe geometries operating at varying resonant frequencies. All measurements were conducted using NaCl solution samples with identical conductivity variations.

RF Probe Design	Operating Frequency, f_0 (MHz)	Slope, R_m/σ (ΩmS^{-1})
Solenoid	10	0.240 ± 0.003
LGR	90	0.20 ± 0.01
Surface coil	50	0.30 ± 0.01

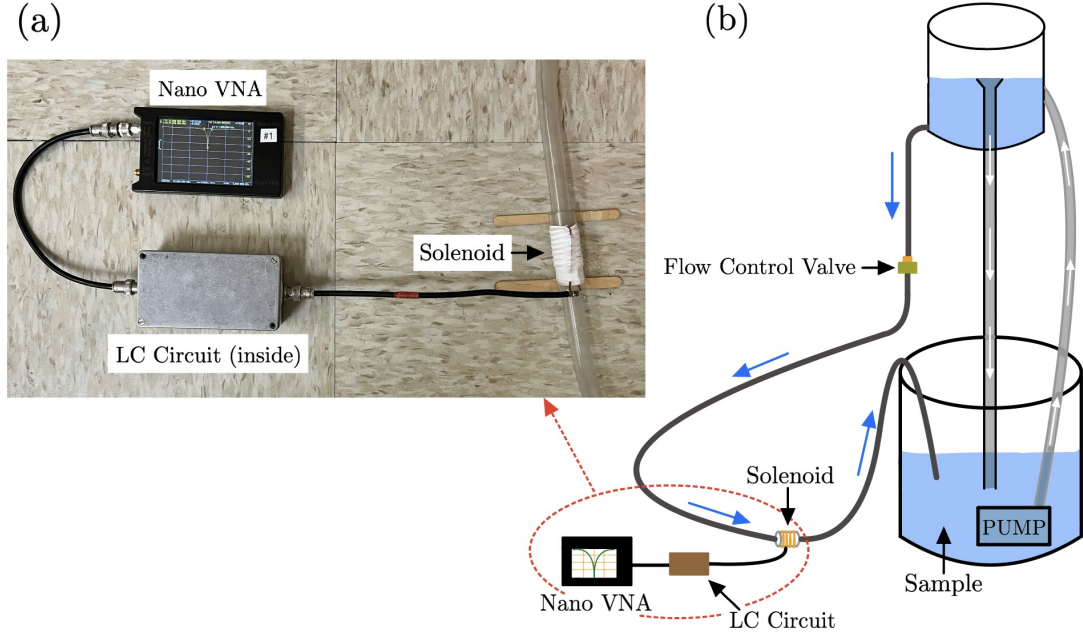


Figure 4.9: Experimental setup. (a) A sample flows through a pipe within a solenoid coil, coupled to an LC circuit in a tuning box connected to a NanoVNA. (b) Schematic of the gravity-fed flow system driving the sample through the coil.

4.2 Exploratory Measurements

This section presents exploratory experiments to evaluate the versatility of our RF probe loading method in two distinct scenarios: measuring the conductivity of flowing aqueous samples and assessing Q-factor variations in common food samples with varying ionic content. While preliminary, these studies aim to demonstrate the method's potential for practical applications, identifying key challenges and standardization needs for future development.

4.2.1 Conductivity Measurement of Flowing Samples

The electrical conductivity of an ionic fluid, such as a salt solution, is governed by the concentration and mobility of its charge-carrying ions, properties that remain invariant under uniform flow relative to a stationary state, provided temperature and composition are constant [32]. This invariance holds because conductivity arises from microscopic ionic motion, which is unaffected by macroscopic fluid dynamics under typical laminar conditions [33]. However, measuring conductivity in flowing samples poses challenges due to potential artifacts – such as mixing, electrode polarization, or flow-induced noise – that may distort traditional contact-based methods. This motivates testing whether our non-invasive RF probe loading technique maintains accuracy in dynamic conditions comparable to static ones. If validated across varying flow rates, this method could enable real-time conductivity monitoring in fluid systems, surpassing the limitations of conventional probes. Here, we assess the solenoidal RF probe’s performance in measuring the conductivity of a salt solution at controlled flow rates with minimal turbulence, using static measurements as a reference.

Results

A solenoid with eight turns, 3.5 cm in length, and a radius of 2.24 cm was wound from copper wire, yielding an inductance of $0.76 \pm 0.04 \mu\text{H}$. Integrated into an LC circuit tuned to 24 MHz, it included fixed capacitors of 15 pF (matching) and 20 pF (tuning) within a tuning box. The sample flowed through a Fisherbrand pipe (inner radius ≈ 2 cm) centered in the solenoid, driven by a gravity-fed system (Fig. 4.9b) with a valve to regulate flow rates from 0 to 3 liters per minute (LPM). Due to volume constraints and the exploratory nature of this study, we tested two samples: tap water (baseline) and a 2% NaCl solution in tap water (saline test case). A portable handheld vector network analyzer model NanoVNA-H (Seesii, Xincheng),

replacing the bulkier 8714B RF Network Analyzer, measured the circuit’s Q -factor.

The experimental procedure was straightforward. Flow rates were determined by timing the filling of a 500 mL cylinder. To ensure laminar flow conditions, we maintained flow rates from 0 to 3 LPM through a 4 cm diameter pipe, yielding Reynolds numbers of 0 to 1592 – well below the turbulent threshold of 2000 – for both tap water and the 2% saline solution at approximately 20°C. The unloaded solenoid’s Q -factor was recorded prior to sample introduction. For each sample – tap water and 2% saline – Q was measured at each flow rate.

Table 4.7 presents the inductive losses (R_m) for tap water and a 2% saline solution at varying flow rates. This flow-rate independence aligns with the hypothesis that conductivity is unaffected by flow, suggesting the RF probe accurately captures sample properties in dynamic conditions. Compared to contact-based meters, which struggle with flowing samples, this non-invasive approach offers a practical advantage for real-time applications.

Table 4.7: Inductive losses (R_m) and Q -factors for tap water and 2% saline solution at varying flow rates, compared to the unloaded probe. The uncertainty in $\Delta f = 0.001$ MHz.

Flow Rate (LPM)	Tap Water			2% Saline		
	Δf (MHz)	Q	R_m ($\pm 0.03 \Omega$)	Δf (MHz)	Q	R_m ($\pm 0.1 \Omega$)
Unloaded	0.30	160 $\pm$ 1	...	0.30	160 $\pm$ 1	...
0	0.42	114.3 $\pm$ 0.4	0.60	0.69	69.6 $\pm$ 0.1	2.0
0.5	0.42	114.3 $\pm$ 0.4	0.60	0.69	69.6 $\pm$ 0.1	2.0
1.8	0.42	114.3 $\pm$ 0.4	0.60	0.69	69.6 $\pm$ 0.1	2.0
3.0	0.42	114.3 $\pm$ 0.4	0.60	0.69	69.6 $\pm$ 0.1	2.0

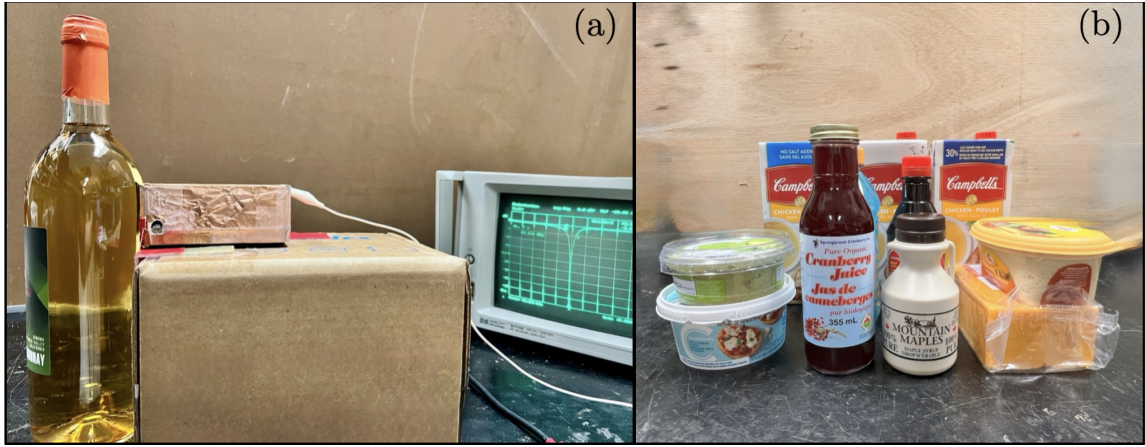


Figure 4.10: (a) Experimental setup with a Chardonnay wine sample positioned near the surface coil RF probe, alongside a VNA displaying real-time frequency response curve. (b) Array of common food samples, including chicken broths (no salt, 30% salt, regular), cranberry juice, maple syrup, guacamole, and cheddar cheese, used for Q -factor change assessment.

4.2.2 Q -Factor Changes in Common Food Samples

To extend the applicability of RF probe techniques beyond controlled saline solutions, we explored conductivity-related losses in a variety of common food samples using our surface coil RF probe tuned to 50 MHz. Unlike the solenoidal probe, which encloses the sample, the surface coil measures conductivity via proximity-induced electromagnetic field perturbations, offering flexibility for samples without compromising geometry. This exploratory study assesses how ionic content and material properties influence the probe’s Q -factor, providing insights into potential non-invasive sensing applications.

Results

Table 4.8 presents preliminary measurements of Δf and Q -factors for a diverse set of food samples—ranging from water-based fluids (e.g., soup broths, juices, wines) to semi-solid matrices (e.g., cheeses, hummus)—using a surface coil RF probe. The Q -factor changes observed here indicate sensitivity to ionic conductivity, though the method’s full applicability across all sample types requires further investigation.

Table 4.8: Measured frequency shifts, Δf and Q-factors for various common food samples using surface coil RF probe using a surface coil RF probe with $L = 0.34 \pm 0.04 \mu\text{H}$ and $f_0 = 50.000 \pm 0.001 \text{ MHz}$.

Sample	Δf ($\pm 0.001 \text{ MHz}$)	Q-factor
Empty Probe	0.692	144.5 ± 0.3
Chicken broth (no salt)	1.122	89.1 ± 0.1
Chicken broth (30% salt)	0.972	102.9 ± 0.2
Chicken broth (regular)	0.921	108.6 ± 0.2
Cranberry Juice	0.768	130.2 ± 0.2
Gatorlyte	0.832	120.2 ± 0.2
Beef Bouillon	1.357	73.7 ± 0.1
Maple Syrup	0.741	135.0 ± 0.3
Hummus	0.793	126.1 ± 0.2
Light Cheddar Cheese	0.877	114.0 ± 0.2
Guacamole	0.953	105.0 ± 0.2
Feta Cheese	1.161	86.1 ± 0.1
Crystalline NaCl	0.707	141.4 ± 0.3
Tempranillo Shiraz (original bottle)	0.792	126.3 ± 0.2
Tempranillo Shiraz (planar sample)	0.784	127.6 ± 0.2
Chardonnay (original bottle)	0.826	121.1 ± 0.2
Chardonnay (planar sample)	0.790	126.6 ± 0.2

Chicken broths were selected to probe salinity effects, with samples labeled “no salt,” “regular,” and “30% salt” expected to show a clear trend in Q -factor reduction with increasing salt content. Contrary to this, the “no salt” broth exhibited the lowest $Q = 89.1$, while the “30% salt” broth yielded $Q = 103.0$, and “regular” broth $Q = 108.6$. Ingredient check revealed that “no salt” broth, despite lower sodium, had higher potassium content, whereas “30% salt” was sodium-dominated. Given potassium’s higher ionic conductivity [33], this explains the observed Q -factor trend, highlighting the probe’s sensitivity to total ionic conductivity rather than specific chemical species.

Fluid samples—cranberry juice, Gatorlyte, and maple syrup—were first measured in their original bottles, then transferred to a standardized planar bottle to minimize container effects on RF probe loading. Variations in container thickness could alter the filling factor (η), the fraction of sample volume coupling with the coil’s magnetic field, affecting Q -factor measurements (§ 4.1.3) [12]. Yet, Q -values stayed consistent across both setups, implying minimal bottle impact for these samples. In contrast, wines—Tempranillo Shiraz (red) and Chardonnay (white)—exhibited bottle-dependent Q -variations. Original bottles gave $Q = 126.3$ (Shiraz) and 121.1 (Chardonnay), differing slightly, but planar bottles aligned them at $Q = 127.6$ and 126.6. This suggests container geometry subtly shifts η , requiring standardized conditions for accurate comparisons.

Semi-solid samples (e.g., feta cheese, guacamole, hummus) produced notable Q -shifts, likely due to their charged ionic content. The Q of the crystalline NaCl sample ($Q = 141.4$) being very close to that of the unloaded probe ($Q = 144.5$) suggests the expected insensitivity of this method to solid salts. These findings underscore the RF probe’s ability to detect ionic conductivity variations across complex matrices, yet highlight the critical need for standardized measurement conditions and calibration—particularly for non-fluid samples—to ensure accuracy and avoid scrutiny, a key consideration for future work.

Chapter 5

Conclusion & Future Work

5.1 Conclusion

This study has successfully demonstrated the viability of a non-invasive, non-contact method for measuring electrical conductivity in water-based samples through RF probe loading, leveraging the inductive losses induced by eddy currents. By analyzing the quality factor response of RF probes to varying sample conductivities, we established a direct correlation between inductive losses, R_m and conductivity, σ , as predicted by our analytical expressions. Theoretical derivations, grounded in classical electromagnetism and reciprocity, provided a solid foundation, with the surface coil analysis employing the magnetic vector potential to address non-uniform fields, yielding results within 1% of established approximations.

Experiments with solenoidal probes, a loop-gap resonator (LGR), and a surface coil across cylindrical and planar samples validated this proportionality, aligning closely with theoretical expectations, despite minor deviations attributable to field distribution and sample volume effects. The consistency of R_m across NaCl and CaCl₂ solutions confirmed that inductive losses depend primarily on conductivity rather than ion type, enhancing the method's robustness. Exploratory tests with flowing samples showed flow-rate independence under laminar conditions, suggest-

ing potential for real-time monitoring, while measurements with common food samples highlighted sensitivity to ionic content, albeit with challenges in heterogeneous matrices requiring standardized conditions. Using a NanoVNA, our setup proved remarkably cost-effective at approximately \$300 CAD.

While not without limitations—such as sensitivity to probe geometry and the need for retuning with sample changes—this technique offers a compelling alternative to traditional contact-based methods, avoiding issues like electrode degradation. It is my hope that this work serves as a valuable resource, laying the groundwork for refining RF probe optimization and expanding its applications in material characterization and biological studies.

5.2 Future Work

Our measurements, with each frequency sweep completed in a fraction of second, are exceptionally fast, though currently constrained by manual tuning and matching for each sample. Future work will explore auto-tuning and matching solutions to streamline the process, boosting efficiency. Building on these results, we aim to refine and broaden this non-invasive technique, potentially integrating it with magnetic resonance. Expanding the frequency range and optimizing probe designs could further improve sensitivity and selectivity. With continued development, RF probe loading has the potential to complement and, in some cases, surpass conventional conductivity measurement techniques, providing a reliable, non-invasive alternative for conductivity sensing in diverse applications.

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Appendix A

Derivation of Inductive Losses in Conductive Samples

The derivation of analytical expressions for inductive losses in conductive samples was primarily based on standard electrodynamics textbooks, including [8, 9]. Many of the equations follow well-established formulations and, therefore, citations are not explicitly provided throughout. This work was carried out during the summer of 2024 and, while not included in the main body of the thesis, it forms the fundamental basis for the research, warranting its inclusion in this appendix.

A.1 Solenoidal RF Probes

Hoult and Lauterbur [5] demonstrated that the noise in NMR experiments with conductive samples arises not only from the resistance of the receiving coil but also from dielectric and inductive losses within the sample. While proper shielding and grounding [7] can minimize dielectric losses, there are no effective methods for mitigating magnetic losses. Therefore, it is important to estimate the dependence of magnetic losses on physical properties of the sample and RF frequency, as well as to measure their values. This section presents the derivation of inductive losses in spherical and cylindrical samples using a solenoidal probe.

A.1.1 Spherical Sample

Consider a sphere of brine solution, modeled as a uniformly conducting sphere, placed within a solenoidal radio-frequency (RF) probe, as illustrated in Fig. 2.3a. This configuration serves as a model to calculate the power loss due to eddy currents induced in the sample. A unit alternating current flows through the coil, producing a magnetic field $\mathbf{B} = B_0 \cos(\omega_0 t) \hat{z}$, assumed homogeneous and uniform across the sample, where B_0 is the field amplitude and ω_0 is the angular frequency. We neglect the skin effect, assuming the skin depth $\delta = (2/(\omega_0 \mu_0 \sigma))^{1/2}$ exceeds the sample radius b , a condition valid at low frequencies [9].

The sphere has radius b and uniform conductivity σ . Eddy currents arise from the time-varying magnetic field, dissipating energy as heat. The power loss P is the Joule heating integrated over the sphere's volume V

$$P = \int_V \sigma |\mathbf{E}|^2 dV, \quad (\text{A.1})$$

where $\mathbf{E}$ is the induced electric field. To find $\mathbf{E}$, we apply Faraday's law in differential form, $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$. For a uniform magnetic field $\mathbf{B} = B_0 \cos(\omega_0 t) \hat{z}$, the time derivative is

$$\frac{\partial \mathbf{B}}{\partial t} = -\omega_0 B_0 \sin(\omega_0 t) \hat{z}. \quad (\text{A.2})$$

In spherical coordinates (r, θ, ϕ) , with $\mathbf{B}$ along the z -axis, symmetry suggests $\mathbf{E}$ is azimuthal ($\hat{\phi}$ -directed) and depends on r and θ . Solving $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ in the low-frequency limit (where the internal field approximates the external field), the induced electric field is [34]

$$\mathbf{E} = \frac{1}{2} \omega_0 B_0 r \sin \theta \sin(\omega_0 t) \hat{\phi}, \quad (\text{A.3})$$

where $r \sin \theta$ is the perpendicular distance from the z -axis. The magnitude squared,

time-dependent, is

$$|\mathbf{E}|^2 = \frac{1}{4} \omega_0^2 B_0^2 r^2 \sin^2 \theta \sin^2(\omega_0 t). \quad (\text{A.4})$$

Substituting into Eq. (A.1), the instantaneous power is

$$\begin{aligned} P &= \sigma \int_V \frac{1}{4} \omega_0^2 B_0^2 r^2 \sin^2 \theta \sin^2(\omega_0 t) dV \\ &= \frac{\sigma \omega_0^2 B_0^2 \sin^2(\omega_0 t)}{4} \int_0^b \int_0^\pi \int_0^{2\pi} r^2 \sin^2 \theta \cdot r^2 \sin \theta d\phi d\theta dr, \end{aligned} \quad (\text{A.5})$$

using the volume element $dV = r^2 \sin \theta dr d\theta d\phi$. Evaluating the integrals separately

$$\int_0^{2\pi} d\phi = 2\pi, \quad \int_0^\pi \sin^3 \theta d\theta = \frac{4}{3} \quad \text{and} \quad \int_0^b r^4 dr = \frac{b^5}{5} \quad (\text{A.6})$$

Thus,

$$P = \frac{\sigma \omega_0^2 B_0^2 \sin^2(\omega_0 t)}{4} \cdot 2\pi \cdot \frac{4}{3} \cdot \frac{b^5}{5} = \frac{2\pi \sigma \omega_0^2 B_0^2 b^5}{15} \sin^2(\omega_0 t). \quad (\text{A.7})$$

The time-averaged power W over a cycle is

$$W = \langle P \rangle = \frac{2\pi \sigma \omega_0^2 B_0^2 b^5}{15} \langle \sin^2(\omega_0 t) \rangle = \frac{\pi \sigma \omega_0^2 B_0^2 b^5}{15}, \quad (\text{A.8})$$

since $\langle \sin^2(\omega_0 t) \rangle = 1/2$. This matches standard results for low-frequency eddy current losses in a conducting sphere [5, 34].

The magnetic field amplitude B_0 at the coil's center, for n turns, radius a , and half-length g , is, via the Biot-Savart law for a long solenoid ($n \gg 1$) [8]

$$B_0 = \frac{n\mu_0 I}{2\sqrt{a^2 + g^2}}, \quad (\text{A.9})$$

with $I = 1$ A (unit current). Substituting into Eq. (A.8)

$$W = \frac{\pi\sigma\omega_0^2 b^5}{15} \left(\frac{n\mu_0}{2\sqrt{a^2 + g^2}} \right)^2 = \frac{\pi\sigma\omega_0^2 \mu_0^2 n^2 b^5}{60(a^2 + g^2)}. \quad (\text{A.10})$$

The effective resistance R_m in series with the coil, where $W = R_m I^2 / 2$ (peak current convention), is

$$R_m = \frac{\pi\omega_0^2 \mu_0^2 n^2 b^5 \sigma}{30(a^2 + g^2)}. \quad (\text{A.11})$$

This derivation agrees with analyses of inductive losses in MRI samples [5] and classical electromagnetic theory [8, 34].

A.1.2 Cylindrical Sample

Consider a homogeneous conducting cylindrical sample within a solenoidal coil, as shown in Fig. 2.3b. Given the similarity to the spherical sample derivation in Section A.1.1, we keep details brief. The cylinder has radius b , length $2g$, and conductivity σ . A unit alternating current in the coil produces a uniform magnetic field $\mathbf{B} = B_0 \cos(\omega_0 t) \hat{z}$, with B_0 as the amplitude and ω_0 the angular frequency, aligned along the cylinder's axis.

The instantaneous power loss P due to eddy currents is the Joule heating integrated over the cylinder's volume V

$$P = \int_V \sigma |\mathbf{E}|^2 dV, \quad (\text{A.12})$$

where $\mathbf{E}$ is the induced electric field. As in Eq. (A.3), for a uniform $\mathbf{B}$ along the z -axis, $\mathbf{E} = \frac{1}{2}\omega_0 B_0 r \sin(\omega_0 t) \hat{\phi}$ in cylindrical coordinates (r, ϕ, z) [34], with r as the radial distance from the axis. Thus,

$$|\mathbf{E}|^2 = \frac{1}{4}\omega_0^2 B_0^2 r^2 \sin^2(\omega_0 t). \quad (\text{A.13})$$

The volume element is $dV = r dr d\phi dz$, and the limits are $r = 0$ to b , $\phi = 0$ to 2π ,

$z = -g$ to g . The power is

$$\begin{aligned} P &= \sigma \int_{-g}^g \int_0^{2\pi} \int_0^b \frac{1}{4} \omega_0^2 B_0^2 r^2 \sin^2(\omega_0 t) \cdot r \, dr \, d\phi \, dz \\ &= \frac{\sigma \omega_0^2 B_0^2 \sin^2(\omega_0 t)}{4} \cdot (2g) \cdot (2\pi) \cdot \int_0^b r^3 \, dr = \frac{\pi \sigma \omega_0^2 B_0^2 b^4 g}{4} \sin^2(\omega_0 t). \end{aligned}$$

The time-averaged power W is

$$W = \langle P \rangle = \frac{\pi \sigma \omega_0^2 B_0^2 b^4 g}{4} \langle \sin^2(\omega_0 t) \rangle = \frac{\pi \sigma \omega_0^2 B_0^2 b^4 g}{8}. \quad (\text{A.14})$$

Using $B_0 = \frac{n\mu_0}{2\sqrt{a^2+g^2}}$ from Eq. (A.9) (unit current), substitute into Eq. (A.14)

$$W = \frac{\pi \sigma \omega_0^2 b^4 g}{8} \left(\frac{n\mu_0}{2\sqrt{a^2+g^2}} \right)^2 = \frac{\pi \sigma \omega_0^2 \mu_0^2 n^2 b^4 g}{32(a^2+g^2)}. \quad (\text{A.15})$$

The effective resistance R_m , where $W = R_m I^2 / 2$ (peak current convention), is

$$\boxed{R_m = \frac{\pi \omega_0^2 \mu_0^2 n^2 b^4 g \sigma}{16(a^2+g^2)}}, \quad (\text{A.16})$$

where a and g are the coil's radius and half-length, and n is the number of turns.

This agrees with Gadian and Robinson [16] and Cho [17], whose results, adjusted for notation, match Eq. (A.16).

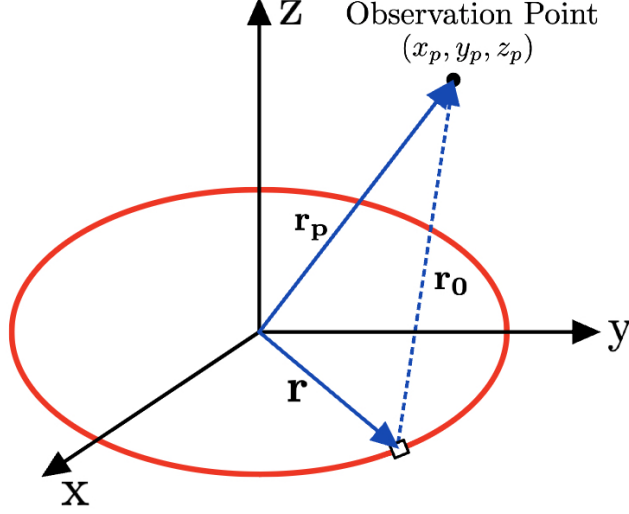


Figure A.1: Circular surface coil positioned in the xy -plane, centered at the origin of the coordinate system. A conducting medium occupies all space for positive z . This configuration is used to calculate the sensitivity, $\mathbf{B}$, and the magnetic vector potential, $\mathbf{A}$, at an observation point $\mathbf{r}_p$ generated by an infinitesimal segment of the coil located at $\mathbf{r}$.

A.2 Surface Coil RF Probes with Planar Samples

In this section, I investigated sample resistance for a surface coil probe with an effectively infinite planar sample, building on our earlier research involving a solenoidal probe. Here, the magnetic field is not treated explicitly; instead, the magnetic vector potential is used to calculate the time-varying electric field induced in the sample by the B -field [8]. The scenario parallels those considered by Hoult and Axel [18], as depicted in Fig. A.1: a round, flat coil positioned in the xy -plane, with the conductive sample occupying the positive z -space and empty space for negative z .

A brief summary of the derivation is provided here to facilitate understanding. We assume a unit alternating current at the resonant frequency flowing through the circular coil and calculate the corresponding magnetic vector potential ($\mathbf{A}$) in space. This employs a quasi-static approach, straightforward for surface coil analysis with sample-induced effects [9]. $\mathbf{A}$ is determined using surface integrals over the area enclosed by the coil [35].

From $\mathbf{A}$, the electric field in the medium is derived, along with the induced current density, related to the electric field via the sample's conductivity. The power dissipation per unit volume is obtained by multiplying the current density by the electric field [34]. Integrating this over the medium's volume yields the total power dissipation. Dividing by the square of the RMS current in the coil gives the sample resistance, R_m [5]. The derivations are extensive, including detailed steps typically omitted, but presented here for completeness and future reference.

Derivation

In gauges where the electric scalar potential ϕ is zero (e.g., the temporal gauge), the electric field $\mathbf{E}$ is expressed as [8]

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad (\text{A.17})$$

where $\mathbf{A}$ is the magnetic vector potential at point $\mathbf{r}_p = (x_p, y_p, z_p)$ induced by unit current I . This simplifies the relationship by focusing on time-varying magnetic effects, neglecting static electric contributions. The electric field $\mathbf{E}$ generates a current density $\mathbf{J}$ in a medium with conductivity σ [9]

$$\mathbf{J} = \sigma \mathbf{E}. \quad (\text{A.18})$$

The time-averaged power P dissipated within a small volume dV is [34]

$$dP = \frac{1}{2} \mathbf{J}^* \cdot \mathbf{E} dV. \quad (\text{A.19})$$

Expressing $\mathbf{A}$ as a complex instantaneous value, $\mathbf{A} = A_0 e^{-i\omega_0 t}$ [8], and applying Eqs. (A.17)–(A.19), the total power dissipation in the medium becomes

$$P = \frac{1}{2} \omega_0^2 \sigma \int_{z>0} \mathbf{A} \cdot \mathbf{A}^* dV, \quad (\text{A.20})$$

where the integral extends over the half-space $V_{1/2}$ ($z > 0$). Since we consider current amplitude rather than RMS, $P = R_m/2$ [5]. Thus, the sample resistance R_m is

$$R_m = \omega_0^2 \sigma \int_{\infty} |\mathbf{A}|^2 dV, \quad (\text{A.21})$$

where $\int_{\infty}$ denotes a 2-D integral over the entire (x_1, y_1) or (x_2, y_2) plane, and I is unit current. The terms sample resistance, inductive loss, and magnetically induced loss are used interchangeably for R_m . In the quasi-static case, the magnetic vector potential for n loops is [9]

$$\mathbf{A} = \frac{\mu_0 n I}{4\pi} \oint \frac{d\mathbf{l}(\mathbf{r})}{|\mathbf{r}_p - \mathbf{r}|}, \quad (\text{A.22})$$

where μ_0 is the permeability of free space, and $d\mathbf{l}$ is the differential length at position $\mathbf{r}$. Numerical integration of Eq. (A.21) using Eq. (A.22) is tedious and prone to singularities near the coil conductors. Instead, $\mathbf{A}$ can be expressed as a surface integral by summing contributions from infinitesimal coils (magnetic dipoles) over the coil's area [35].

To express $\mathbf{A}$ as a surface integral, the coil's area is divided into infinitesimal coils (magnetic dipoles) with dipole moments $I ds$, as shown in Fig. 2.2 b. The total field is the superposition of these contributions [36]. Per Eq. (A.22), an infinitesimal coil at $\mathbf{r} = (x, y)$ in the $z = 0$ plane, with current I and area ds , generates a vector potential $\mathbf{a}(x, y) ds$ at $\mathbf{r}_p = (x_p, y_p, z_p)$

$$\begin{aligned} \mathbf{a}(x, y) ds &= \frac{\mu_0 I}{4\pi} \nabla_p \times \frac{\mathbf{e}_z}{|\mathbf{r}_p - \mathbf{r}|} ds \\ &= \frac{\mu_0 I}{4\pi} \frac{(y - y_p)\mathbf{e}_x + (x - x_p)\mathbf{e}_y}{r_0^3} ds, \end{aligned} \quad (\text{A.23})$$

where $r_0 = |\mathbf{r}_p - \mathbf{r}| = \sqrt{(x_p - x)^2 + (y_p - y)^2 + z^2}$. The total vector potential is

$$\mathbf{A}(x, y) = \int_S \mathbf{a}(x, y) ds, \quad (\text{A.24})$$

where S is the coil's area. To generalize this formulation, a weighting function $f(x, y)$ is introduced. For a single coil, this function is defined as 1 within the area enclosed by the coil and 0 outside it

$$f(x, y) = \begin{cases} 1 & \text{within the coil area,} \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.25})$$

The function $f(x, y)$ is used in the calculation of the magnetic field and noise resistance by effectively restricting the integration domain to the area inside the coil. With this weighting function, the vector potential $\mathbf{A}$ can now be expressed as an infinite surface integral extending over the entire $z = 0$ plane

$$\mathbf{A}(x, y) = \int_{\infty} \mathbf{a}(x, y) f(x, y) ds. \quad (\text{A.26})$$

Using Eqs. (A.23) and (A.26), the integrand in Eq. (A.21) expands as

$$\begin{aligned} |\mathbf{A}|^2 &= |\mathbf{A} \cdot \mathbf{e}_x|^2 + |\mathbf{A} \cdot \mathbf{e}_y|^2 = \left| \int_{\infty} a_x(x, y) f(x, y) ds \right|^2 + \left| \int_{\infty} a_y(x, y) f(x, y) ds \right|^2 \\ &= \int_{\infty} \int_{\infty} [a_x(x_1, y_1) a_x^*(x_2, y_2) + a_y(x_1, y_1) a_y^*(x_2, y_2)] \cdot f(x_1, y_1) f^*(x_2, y_2) ds_1 ds_2 \\ &= \left(\frac{\mu_0 \mathbf{I}}{4\pi} \right)^2 \int_{\infty} \int_{\infty} \frac{(x_p - x_1)(x_p - x_2) + (y_p - y_1)(y_p - y_2)}{r_1^3 r_2^3} \cdot f(x_1, y_1) f^*(x_2, y_2) ds_1 ds_2, \end{aligned} \quad (\text{A.27})$$

where

$$\begin{aligned} r_1 &= \sqrt{(x_p - x_1)^2 + (y_p - y_1)^2 + z_p^2}, \\ r_2 &= \sqrt{(x_p - x_2)^2 + (y_p - y_2)^2 + z_p^2}, \end{aligned}$$

and $\int_{\infty} ds_1, \int_{\infty} ds_2$ represent 2D integrals over the entire (x_1, y_1) and (x_2, y_2) planes, respectively. (Note that $\mathbf{A}$ has no z -component.) Combining with Eq. (A.21), the loss resistance R_m can be expressed as

$$R_m = \omega_0^2 \sigma \left(\frac{\mu_0 n I}{4\pi} \right)^2 \int_{\infty} \int_{\infty} \xi(x_1, x_2, y_1, y_2) f(x_1, y_1) f^*(x_2, y_2) ds_1 ds_2, \quad (\text{A.28})$$

where $\xi(x_1, x_2, y_1, y_2)$ is defined as

$$\xi(x_1, x_2, y_1, y_2) = \int_{z_p > 0} \frac{(x_p - x_1)(x_p - x_2) + (y_p - y_1)(y_p - y_2)}{r_1^3 r_2^3} dV, \quad (\text{A.29})$$

with the integral $\int_{z_p > 0} dV$ denoting a volume integral over the half-space $z > 0$. Due to the symmetry of the problem, it is evident that this function depends solely on the distance d between the points (x_1, y_1) and (x_2, y_2)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \quad (\text{A.30})$$

Setting $x_1 = d, y_1 = 0, x_2 = 0, y_2 = 0$, the expression for ξ simplifies to

$$\xi = \int_{z_p > 0} \frac{(x_p - d)x_p + y_p^2}{[(x_p - d)^2 + y_p^2 + z_p^2](x_p^2 + y_p^2 + z_p^2)^{3/2}} dV. \quad (\text{A.31})$$

To simplify the calculation, the integral is extended to the entire space

$$\xi = \frac{1}{2} \int_{\infty} \frac{(x_p - d)x_p + y_p^2}{[(x_p - d)^2 + y_p^2 + z_p^2](x_p^2 + y_p^2 + z_p^2)^{3/2}} dV. \quad (\text{A.32})$$

In cylindrical coordinates $x_p = x_p, y_p = t \cos \theta, z_p = t \sin \theta$

$$\xi = \frac{\pi}{2} \int_{-\infty}^{\infty} \int_0^{\infty} \frac{2t(x_p - d)x_p + t^3}{[(x_p - d)^2 + t^2](x_p^2 + t^2)^{3/2}} dt dx_p, \quad (\text{A.33})$$

with the inner integral yielding

$$\int_0^\infty \frac{2t(x_p - d)x_p + t^3}{[(x_p - d)^2 + t^2](x_p^2 + t^2)^{3/2}} dt = \begin{cases} -\frac{1}{d^2}, & 0 < x_p < d, \\ \frac{3}{(2x_p - d)^2}, & \text{otherwise.} \end{cases} \quad (\text{A.34})$$

Thus, ξ simplifies to

$$\xi = \frac{\pi}{2} \left(\int_{-\infty}^0 \frac{3}{(2x_p - d)^2} dx_p - \int_0^d \frac{1}{d^2} dx_p + \int_d^\infty \frac{3}{(2x_p - d)^2} dx_p \right) = \frac{\pi}{d}. \quad (\text{A.35})$$

Defining $\eta = \omega_0^2 \sigma \left(\frac{\mu_0 n I}{4\pi} \right)^2$, Eq. (A.28) becomes:

$$R_m = \eta \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{\pi f(x_1, y_1) f^*(x_2, y_2)}{d} ds_1 ds_2. \quad (\text{A.36})$$

This integral shows that for any arbitrary coil definition function $f(x, y)$, there is no singularity in the calculation of the R_m (even for wires with radius, $a = 0$). By transforming this equation into the Fourier domain, the convolution theorem can be applied and one of the integrals can be replaced with a simple product. Transforming to the Fourier domain, the 2-D Fourier transform of $f(x, y)$ is [37]:

$$F(u, v) = \frac{1}{2\pi} \int_{-\infty}^\infty \int_{-\infty}^\infty f(x, y) e^{-j(ux+vy)} dx dy. \quad (\text{A.37})$$

For $f(x, y) = 1$ within a coil of radius a and 0 outside (Eq. (A.25)), in polar coordinates

$$F(u, v) = \frac{1}{2\pi} \int_0^{2\pi} \int_0^a e^{-jr(u \cos \theta + v \sin \theta)} r dr d\theta = \frac{a J_1(\beta a)}{\beta}, \quad (\text{A.38})$$

where $\beta = \sqrt{u^2 + v^2}$, and J_1 is the Bessel function of the first kind, order 1 [38]. By convolution theorem [37], (A.36) becomes

$$\begin{aligned}
R_m &= 2\pi^2\eta \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{|F(u,v)|^2}{\beta} du dv \\
&= 2\pi^2\eta \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{a^2 J_1^2(\beta a)}{\beta^3} du dv
\end{aligned} \tag{A.39}$$

Given the radial symmetry of the problem, it's convenient to switch to polar coordinates i.e., (u, v) to (β, θ) ; where $u = \beta \cos \theta$, $v = \beta \sin \theta$ and $du dv = \beta d\beta d\theta$. The integral then becomes

$$\begin{aligned}
R_m &= 2\pi^2\eta \int_0^{2\pi} \int_0^{\infty} \frac{a^2 J_1^2(a\beta)}{\beta^3} \cdot \beta d\beta d\theta \\
&= 4\pi^3\eta a^2 \int_0^{\infty} \frac{J_1^2(a\beta)}{\beta^2} d\beta.
\end{aligned} \tag{A.40}$$

Let $u = a\beta$, then $d\beta = du/a$. When $\beta = 0$, $u = 0$, and when $\beta = \infty$, $u = \infty$. Therefore,

$$R_m = 4\pi^3\eta a^3 \int_0^{\infty} \frac{J_1^2(u)}{u^2} du = \frac{16\pi^2}{3}\eta a^3 \tag{A.41}$$

since the integral $\int_0^{\infty} \frac{J_1^2(x)}{x^2} dx = \frac{4}{3\pi}$. With $\eta = \omega_0^2 \sigma \left(\frac{\mu_0 n \mathbf{I}}{4\pi}\right)^2$, the inductive loss R_m for a circular loop surface coil with unit current is then

$$\boxed{R_m = \frac{1}{3}\omega_0^2 \mu_0^2 n^2 a^3 \sigma}, \tag{A.42}$$

where ω_0 is the resonant frequency of the RF probe, μ_0 is the permeability of the free space, n is the number of turns of the coil, a is the radius of the surface coil, and σ is the conductivity of the sample. Harpen [19] approached the problem similarly but used a summation approximation for an integral when calculating the magnetic vector potential $\mathbf{A}$ in spherical coordinates, employing an expansion in spherical harmonics. His approximation resulted in an inductive loss

$$R_m = 0.0332 \pi^2 \omega_0^2 \mu_0^2 n^2 a^3 \sigma, \quad (\text{A.43})$$

which deviates from Eq. (A.42) by a factor of about 1%. In both equations, R_m is proportional to the conductivity σ of the sample and the square of resonant frequency ω_0 of the circuit.

Appendix B

Uncertainty Propagation in Measured Inductive Losses

Uncertainties in the experimental parameters are propagated using the standard uncertainty propagation method outlined in Taylor's *Error Analysis* [39, 40]. The Q-factor is defined as (Eq. (3.3))

$$Q = \frac{2\omega_0}{\Delta\omega} = \frac{2f_0}{\Delta f}, \quad \text{where} \quad \omega_0 = 2\pi f_0 \quad \text{and} \quad \omega = 2\pi f. \quad (\text{B.1})$$

Note that; $\Delta f = f_2 - f_1$. For each frequency measurement, f exhibit uncertainty of approximately 0.001 MHz. Following Taylor's recommendation for independent random errors, the uncertainty in Δf is calculated using quadrature:

$$\delta(\Delta f) = \sqrt{(\delta f_1)^2 + (\delta f_2)^2} = \sqrt{0.001^2 + 0.001^2} \approx 0.001414 \text{ MHz}. \quad (\text{B.2})$$

Then, the uncertainty in the Q-factor is

$$\delta Q = \sqrt{\left(\frac{\partial Q}{\partial f_0} \cdot \delta f_0\right)^2 + \left[\frac{\partial Q}{\partial(\Delta f)} \cdot \delta(\Delta f)\right]^2}$$

$$\delta Q = \frac{2}{\Delta f} \sqrt{(\delta f_0)^2 + \left[\frac{f_0}{\Delta f} \cdot \delta(\Delta f) \right]^2}. \quad (\text{B.3})$$

This δQ applies to both the loaded and unloaded probes, as similar frequency deviations are observed. Recall Eq. (2.5):

$$R_m = \omega_0 L \cdot \Delta \frac{1}{Q}, \quad (\text{B.4})$$

where $\Delta \frac{1}{Q} = \frac{1}{Q_l} - \frac{1}{Q_u}$. The uncertainty in $\Delta(1/Q)$ is then

$$\begin{aligned} \delta \left(\Delta \frac{1}{Q} \right) &= \sqrt{\left[\frac{\partial}{\partial Q_u} \left(\Delta \frac{1}{Q} \right) \cdot \delta Q_u \right]^2 + \left[\frac{\partial}{\partial Q_l} \left(\Delta \frac{1}{Q} \right) \cdot \delta Q_l \right]^2} \\ &= \sqrt{\left(\frac{\delta Q_u}{Q_u^2} \right)^2 + \left(\frac{\delta Q_l}{Q_l^2} \right)^2}. \end{aligned} \quad (\text{B.5})$$

The uncertainty in inductive loss R_m is then:

$$\begin{aligned} \delta R_m &= \sqrt{\left[\frac{\partial R_m}{\partial (\Delta(1/Q))} \cdot \delta \left(\Delta \frac{1}{Q} \right) \right]^2 + \left(\frac{\partial R_m}{\partial L} \cdot \delta L \right)^2} \\ &= \omega_0 \sqrt{\left[L \cdot \delta \left(\Delta \frac{1}{Q} \right) \right]^2 + \left[\left(\Delta \frac{1}{Q} \right) \cdot \delta L \right]^2}. \end{aligned} \quad (\text{B.6})$$

Simplifying this expression by normalizing by R_m , we elegantly obtain the relative uncertainty:

$$\boxed{\delta R_m = R_m \sqrt{\left(\frac{\delta(\Delta(1/Q))}{\Delta(1/Q)} \right)^2 + \left(\frac{\delta L}{L} \right)^2}}, \quad (\text{B.7})$$

which recovers Eq. (B.6) upon substituting Eq. (2.5). As an example, consider the first experiment using a 10 MHz solenoidal RF probe (§ 4.1.1). The inductance L was measured five times, yielding an average of $3.420 \mu\text{H}$. The LC meter's specified accuracy of 1%+1 [41] gives a systematic uncertainty of $\delta L_s = 0.035 \mu\text{H}$, while the re-

peated measurements result in a random uncertainty of $\delta L_r = 0.002 \mu\text{H}$. Combining these in quadrature, the total uncertainty is:

$$\delta L = \sqrt{\delta L_s^2 + \delta L_r^2} = \sqrt{(0.035)^2 + (0.002)^2} \approx 0.035 \mu\text{H}. \quad (\text{B.8})$$

Thus, $L = 3.420 \pm 0.035 \mu\text{H}$. For the empty probe, $Q_u = 204.1$ ($\Delta f = 0.098 \pm 0.001414 \text{ MHz}$), and for the 4% salinity sample, $Q_1 = 85.8$ ($\Delta f = 0.233 \pm 0.001414 \text{ MHz}$). Substituting into Eq. B.3, the uncertainties are:

$$\begin{aligned} \delta Q_u &= \frac{2}{0.098} \sqrt{0.001^2 + \left(\frac{10 \times 0.01414}{0.098}\right)^2} \approx 2.94 \quad \text{and} \\ \delta Q_1 &= \frac{2}{0.228} \sqrt{0.001^2 + \left(\frac{10 \times 0.01414}{0.228}\right)^2} \approx 0.54. \end{aligned} \quad (\text{B.9})$$

From Eq. (B.5), the uncertainty in $\Delta(1/Q)$ is:

$$\delta \left(\Delta \frac{1}{Q} \right) = \sqrt{\left(-\frac{2.94}{204.1^2} \right)^2 + \left(\frac{0.54}{87.7^2} \right)^2} \approx 0.0001. \quad (\text{B.10})$$

Finally, using Eq. (B.7) for the 4% salinity sample ($R_m \approx 1.4 \Omega$), the uncertainty in inductive loss is:

$$\delta R_m = 1.4 \cdot \sqrt{\left(\frac{0.0001}{0.0065} \right)^2 + \left(\frac{0.035}{3.420} \right)^2} \approx 0.026 \Omega, \quad (\text{B.11})$$

where $\Delta(1/Q) = 0.0065 \pm 0.0001$. Hence, the inductive loss for 4% salinity sample is $R_m = 1.40 \pm 0.03 \Omega$. This process is applied to determine uncertainties in inductive losses for all samples and probes. Analyzing and reporting the uncertainties enhances the reliability of the inductive loss measurements, supporting the experimental results in Chapter 4.